Energy Conversion and Management: X 26 (2025) 100910

Contents lists available at ScienceDirect

Energy Conversion and Management: X

journal homepage: www.sciencedirect.com/journal/energy-conversion-and-management-x

Using the Taguchi method and grey relational analysis to optimize the parameter design of flat-plate collectors with nanofluids, and phase change materials in an integrated solar water heating system

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ARTICLE INFO

Keywords:
Solar water heating system
Flat plate collector
Phase change material, Nanofluids
Taguchi method
Grey relational analysis method

ABSTRACT

Solar water heating systems play a critical role in renewable energy applications, with their global market experiencing continuous growth due to increasing recognition and demand. Despite this, such systems face dual challenges in practical use: low thermal storage efficiency and insufficient heat retention. This study introduces an innovative integration of nanofluids and phase change materials (PCMs) with flat-plate collectors, representing a novel approach to addressing these challenges. Unlike prior studies that focused on individual technologies, this research combines advanced materials with multi-objective optimization to significantly enhance system performance. The Taguchi method was employed to plan the experiments, with nine control factors selected: PCM material, PCM volume, number of PCM tubes, working fluid, mass flow rate, number of collector tubes, collector tube material, tilt angle, and azimuth angle. A total of 36 experiments were designed using the Taguchi orthogonal array and simulated through TRNSYS software. Data from these experiments were analyzed using Signal-to-Noise (S/N) ratios, main effects analysis, and analysis of variance to identify optimal parameter combinations. Finally, grey relational analysis was utilized for multi-quality optimization, enabling the simultaneous enhancement of thermal storage efficiency and heat retention time. The results demonstrate that the optimized configuration achieved a thermal storage efficiency of 94.2 % and a heat retention time of 31.7 h. The optimal parameters included the use of PCM material, 20 % PCM volume, 14 PCM tubes, CuO nanofluid as the working fluid, a mass flow rate of 0.02 kg/s, 9 collector tubes, copper collector plates, a tilt angle of 22.4°, and an azimuth angle of 0° facing south. Compared to the non-optimized system, these optimizations increased thermal storage efficiency by 28 % and extended heat retention time by 14.6 h. The innovative integration and optimization framework presented in this study not only bridges the gap between theoretical research and practical applications but also provides a scalable solution for improving the efficiency of renewable energy systems.

1. Introduction

In contemporary society, energy shortages and environmental pollution have become increasingly critical challenges [1]. Facilitating the energy transition and fostering the development of renewable energy are pivotal to achieving sustainable development goals [2]. In Taiwan, the primary renewable energy sources include solar thermal, photovoltaic, and wind energy [3]. Solar thermal systems harness solar energy for heating and hot water supply [3,4], while photovoltaic systems convert sunlight into electricity through solar cells [5]. Wind energy is predominantly utilized for power generation [6]. The Taiwanese government has actively promoted energy diversification and sustainability through supportive policies and investment incentives [7]. According to the latest report by the International Renewable Energy Agency, thermal energy constitutes 48.7 % of the global total energy supply, with solar energy contributing 10.5 % to modern renewable heat production. The global market for solar thermal collectors has experienced a 3 % growth, achieving a cumulative capacity of 522 GW-T [8]. These developments underscore the critical role of solar water heating systems in advancing renewable heat supply. A solar water

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https://doi.org/10.1016/j.ecmx.2025.100910
Received 30 November 2024; Received in revised form 23 January 2025; Accepted 2 February 2025
Available online 6 February 2025

heating system typically comprises three main components: solar collectors, thermal storage tanks, and heat transfer fluids [9,10]. To enhance system performance, auxiliary components such as pumps, electric heaters, and heat exchangers are integrated. These additions ensure efficient operation under varying weather conditions, providing a reliable hot water supply for residential and commercial applications [11].

Significant research efforts have been devoted to studying and optimizing flat-plate solar collectors to enhance overall system performance. A flat-plate solar collector is a device designed to harness solar energy for heat generation [12]. Pioneering work on flat-plate collectors was conducted by Hottel in 1942 [13], introducing a design that incorporated a glass cover, a black absorber plate, insulation, and a casing, establishing the foundation for solar thermal applications.

Subsequent studies have focused on various aspects of flat-plate collector performance. Shariah [14] investigated the effect of collector plate thermal conductivity, demonstrating that aluminum plates outperform steel, while copper plates provide the most substantial improvement. Bello [15] analyzed the influence of collector tube spacing on performance, finding that narrower spacing increases efficiency but with diminishing returns, emphasizing the need for an optimal spacing balance. Islam [16] examined the geometric configurations of collector tubes, revealing that square-shaped tubes significantly enhance efficiency, achieving a maximum of 70.44 %.

Wang [17] investigated the influence of mass flow rate on the performance of solar collectors, concluding that higher flow rates enhance heat exchange efficiency and decrease thermal resistance. However, the study noted that flow rate increases beyond a critical threshold result in negligible performance improvements. Similarly, Moghadam [18] examined the effect of tilt angle on flat-plate collector efficiency, demonstrating that precise adjustments to the tilt angle can significantly improve thermal performance. Zheng [19] designed a flat plate solar collector with plastic and silica aerogel insulation, achieving 50 % higher efficiency than commercial models and reducing heat losses in high-temperature applications. Gong [20] developed a V-corrugated flat plate solar collector (VC-FPSC) with SSD/SAT phase change materials, enhancing thermal performance by extending heating duration and improving stability, especially at night. Li [21] introduced an S-shaped bend tube for flat-plate collectors based on field synergy theory, reducing plate temperature by 0.33°C and improving efficiency by 0.1 %. El-Fakharany [22] designed a solar air heater (SAH) combining flat plate and evacuated tube collectors with phase change materials (PCM), achieving efficiencies of up to 64.53 %, outperforming traditional designs.

Kuo [23] utilized the Taguchi method integrated with grey relational analysis to assess multiple quality characteristics of flat-plate collectors, effectively identifying the optimal process parameters to enhance overall system performance.

Advancements in nanotechnology have led to the development of a novel category of heat transfer fluids known as nanofluids. These are specially engineered fluids containing solid nanoparticles dispersed on the nanoscale [24], which may include metals, oxides, carbon nanotubes, and other materials. In solar collectors, the incorporation of nanofluids enhances radiative heat transfer and thermal conductivity [25], thereby significantly improving efficiency and performance. Otanicar [26] demonstrated that nanofluids composed of carbon nanotubes, graphite, and silver could enhance solar collector efficiency by up to 5 %. Hawwash [27] reported that alumina-based nanofluids increased the thermal efficiency of flat-plate solar collectors by 18 %, outperforming pure water. Similarly, Yousefi [28] found that as the mass flow rate increased, alumina nanofluids improved flat-plate collector efficiency by 28.3 %. Prakasam [29] observed a 14.3 % improvement in collector efficiency at a flow rate of 2 L per minute (L/min) when using alumina nanofluids compared to pure water. Further investigations by Ashour [30] revealed that copper oxide and zinc oxide nanofluids achieved efficiencies of up to 81.64 % and 77.64 %, respectively, under varying flow rates, significantly exceeding the 60.21 % efficiency of pure water. Mirzaei [31] highlighted the influence of flow rates on nanofluid performance, showing that copper oxide enhanced efficiency by 8.2 % at 1 L/min and by 7.9 % at 4 L/min, while alumina improved efficiency by 7.2 % at 1 L/min and by 5.2 % at 4 L/min. These results underscore the substantial benefits of using nanofluids at lower flow rates to maximize thermal efficiency. Desisa [32] conducted a comparative study on the performance of copper oxide, alumina, titanium oxide, and silicon dioxide nanofluids, concluding that copper oxide provided the most pronounced efficiency improvement among the tested nanofluids. Collectively, these studies highlight the considerable potential of nanofluids, particularly those based on alumina and copper oxide, in enhancing solar collector performance. Additionally, their cost-effectiveness relative to other nanoparticles further supports their practical application [33].

As research progresses, phase change materials (PCMs) have garnered significant attention due to their numerous advantages. PCMs possess high latent heat, enabling them to absorb or release substantial amounts of energy within a compact volume through phase change processes [34]. Furthermore, PCMs exhibit appropriate phase change temperatures and favorable thermal conductivity within specific working temperature ranges, ensuring efficient heat transfer and consistent phase change behavior [35]. These properties make PCMs a key component in solar thermal storage systems, where they store heat during the day and release it at night or when required [36]. Several studies have highlighted the efficacy of PCMs in enhancing solar water heating system performance. Cabeza [37] demonstrated that increasing the number and volume of PCM modules in a solar water heating system significantly improved energy density and thermal efficiency. Ferrer [38] evaluated the suitability of different container materials, finding that stainless steel offers excellent corrosion resistance when used with PCMs, whereas aluminum and carbon steel may be prone to corrosion under certain conditions. Hinti [39] investigated the integration of paraffin-based PCM into conventional solar water heating systems. The findings revealed that the inclusion of PCM facilitated greater heat absorption during the day and consistent heat release at night, effectively maintaining water temperature. Kumar [40] designed a solar water heating system utilizing two distinct types of PCMs and compared it with a system lacking PCM. The PCM-integrated system exhibited substantially higher efficiency, with tank water temperatures significantly elevated the following morning compared to the non-PCM system. Zhao [41] explored the performance of organic paraffin as a PCM in storage tanks, showing that although PCM occupied less than 20 % of the tank's volume, it contributed to 50 % of the total stored heat. This utilization not only enhanced energy efficiency but also extended the hot water supply duration during periods without solar radiation. Yousif [42] evaluated a CPC solar air collector with phase change materials (PCM). The normal CPC achieved 77.1 % efficiency at 0.027 kg/s airflow, outperforming the involute design by 13 %, highlighting its suitability for domestic air heating.

The Taguchi method is a widely recognized tool for optimizing engineering processes focused on a single quality characteristic. By employing orthogonal arrays for experimental design, it facilitates data analysis to enhance product quality and determine the optimal combination of control factors and levels. However, in practical scenarios involving multiple quality requirements, it becomes essential to integrate additional multi-quality analysis methods. Grey relational analysis (GRA) serves as a powerful complementary technique due to its capability to evaluate complex relationships among multiple quality indicators and influencing factors. This integration enables comprehensive multi-quality optimization, ensuring the robustness and reliability of the results. For instance, Kazemian [43] applied the Taguchi method in conjunction with grey relational analysis to optimize the operating conditions of a photovoltaic thermal (PV/T) and solar collector combination system. The study systematically analyzed the effects of parameters such as mass flow rate and inlet temperature, successfully improving both power generation and thermal efficiency. Similarly, Liu [44] utilized the Taguchi method and GRA to optimize a PV/T system incorporating organic paraffin and nanofluids, aimed at enhancing power generation and heat storage efficiency in residential solar systems. The experiments demonstrated significant performance improvements, achieving a power generation efficiency of 14.958 % and a heat storage efficiency of 64.764 %.

Multi-response optimization problems can be effectively analyzed using the Taguchi method combined with grey relational analysis. Mrmon et al. [45] evaluated the performance of conventional and hybrid woven fabrics for sustainable protective clothing. Their study identified that a Kevlar and Ramie fabric with an 80:20 ratio and a twill weave delivered the best performance across various tests. Sumesh and Kanthavel [46] optimized tensile, flexural, and impact strength in banana/coir epoxy composites using Taguchi-based grey relational analysis. They found that a combination of 20 % banana, 15 % coir, 3 % alkali treatment, 16 MPa pressure, and 100 °C temperature provided superior mechanical properties. Üstüntağ and Şenyiğit [47] optimized coating process parameters for cotton/elastane denim fabrics, determining the ideal conditions as 14 picks/cm weft density, 160 °C drying temperature, 50 dPa.s viscosity, 3 bar squeeze pressure, and 30 m/min fabric speed. Jogi et al. [48] applied this methodology to optimize machining parameters for PEEK polymer using CNC turning machines, identifying 377 m/min cutting speed, 0.05 mm/rev feed rate, 2 mm depth of cut, and TiC insert as optimal.

Based on the studies [45–48], the grey relational analysis model is generally effective for optimizing multiple responses by converting them into a single relational grade [47]. This approach facilitates the simultaneous evaluation of various objective functions and helps identify the optimal parameters for all-purpose functions [49]. The Taguchi method was initially used to optimize each product quality individually, followed by the integration of all qualities using grey relational analysis while considering the prioritization of objectives [47]. Consequently, the most influential parameters for all qualities were determined systematically.

This study aims to enhance the performance of solar water heating systems by developing an innovative “integrated solar water heating system” that combines nanofluids, phase change materials (PCMs), and flat-plate collectors, with optimized parameter configurations. Despite significant advancements in individual technologies, existing studies often fail to address the critical challenges of combining these advanced materials to achieve simultaneous improvements in thermal storage efficiency and heat retention time. Furthermore, there is limited exploration of systematic multi-objective optimization frameworks that integrate these technologies for practical applications.

By systematically analyzing the interactions among key performance parameters and employing optimization theory, this study identifies the most effective configuration to achieve superior system performance. The primary innovations include the integration of advanced materials with multi-objective optimization techniques to overcome the limitations of individual technologies. This approach not only enhances the thermal performance of solar water heating systems but also bridges the gap between theoretical research and real-world applications. Ultimately, these advancements provide a scalable and sustainable solution for renewable energy systems, addressing unresolved issues in the field and offering practical value for energy-efficient technologies.

2. Integrated solar water heating system structure and software performance testing

The integrated solar water heating system leverages advanced technologies, including nanofluids, phase change materials (PCMs), and flat-plate collectors, to enhance thermal storage efficiency and prolong heat retention time. The system comprises essential components such as solar collectors, thermal storage tanks, and heat transfer fluids, complemented by auxiliary elements like pumps, ensuring efficient operation and reliable performance, as shown in Fig. 1. The detailed specifications of the system module are shown in Table 1.

The flow diagram of the integrated solar water heating system investigated in this study is shown in Fig. 2, and the corresponding physical system is illustrated in Fig. 3.

In this study, the TRNSYS simulation tool was utilized to model and analyze the integrated solar water heating system, aiming to optimize design parameters for improved thermal storage efficiency and extended heat retention duration. During the experimental process, parameters of various modules were continuously adjusted to evaluate the impact of different configurations on system performance. Conducting such modifications with physical modules would have required substantial effort, significantly complicating the experiments. The primary advantage of TRNSYS lies in its capability to simulate the system’s operation under diverse environmental conditions by altering parameters of individual components within the integrated solar water heating system. These parameters include the use of PCM materials, the tilt angle of solar collectors, and the materials of the absorber plates. The modeling and connection methodology employed in this experiment is depicted in Fig. 4.

3. Research methodology

To minimize the number of experiments and associated costs, this study employed the Taguchi method for experimental design. While the Taguchi method is well-suited for process optimization focusing on a single quality characteristic, the objectives of this research involve optimizing multiple quality characteristics of the system. Consequently, the Taguchi method was combined with grey relational analysis to address this complexity. This integration enabled the optimization of parameter combinations and facilitated decision analysis for multiple quality objectives, aiming to achieve comprehensive optimization across all targeted quality characteristics.

3.1. Taguchi method

The Taguchi method was employed to design the experimental parameters, including control factors and their respective levels. Main Effects Analysis (MEA) and Analysis of Variance (ANOVA) were subsequently used to assess the influence of product parameters on the quality characteristics. The Signal-to-Noise (S/N) ratio served as a key indicator for evaluating the manufacturing process and quality levels. Depending on the type of quality characteristic being analyzed, the S/N ratio can be calculated using one of four methods: “Larger-the-Better,” “Smaller-the-Better,” “Nominal-the-Best,” or “Target-the-Best.” As this study focuses on maximizing thermal storage efficiency and heat retention time, the “Larger-the-Better” method was applied for calculating the S/N ratio.

$$S/N_{LTB} = -10 \bullet \log_{10}\left(\frac{1}{n}\sum_{i=1}^{n}\frac{1}{{y_i}^2}\right) \quad (1)$$

where y_i represents the experimental data, and n is the number of measurements.

3.2. Main effects analysis

Main Effects Analysis (MEA) evaluates the influence of factors on system performance across different levels. Based on the experimental design requirements, the S/N ratio for each experiment is first calculated. Subsequently, the average response value Fi for each factor at each level is computed, followed by determining the main effect variation ΔF for each factor across its levels. These values are then organized into a response table for the analysis of main effects.

Assuming a factor has i levels, with mmm sets of S/N ratios for each level, the formulas for the average response value Fi and the main effect variation ΔF are as follows:

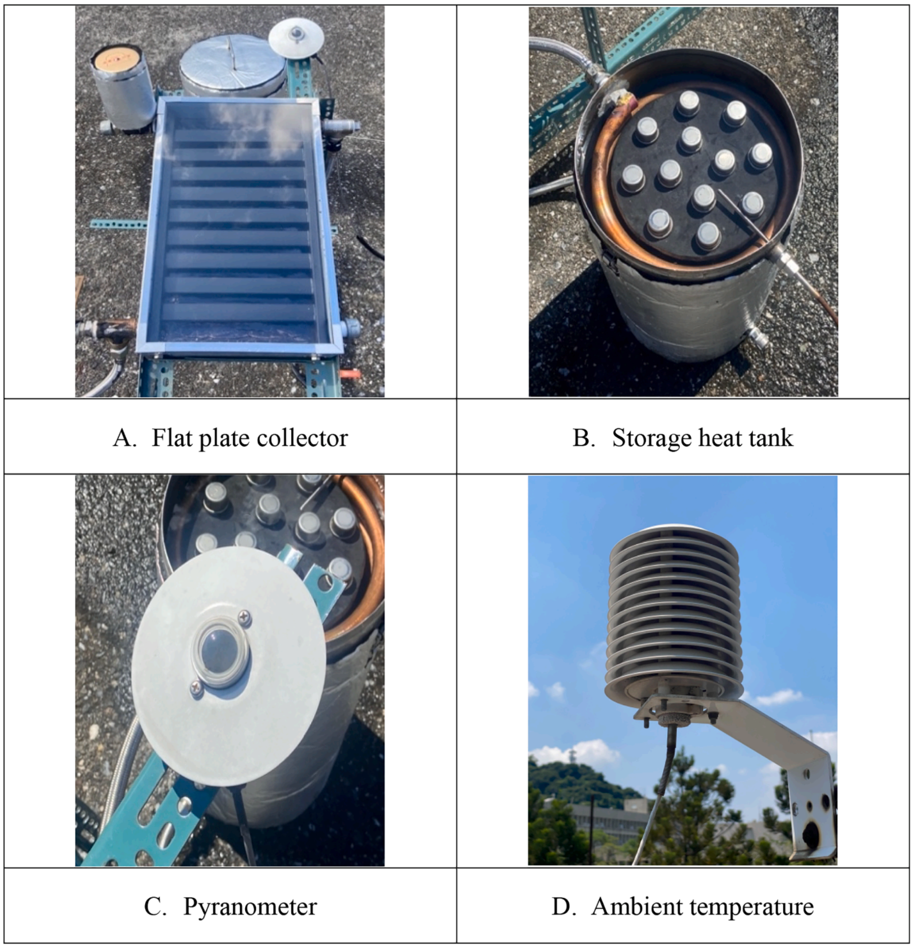


Fig. 1. Diagram of physical equipment and sensors.

$$F_i = \frac{1}{m}\sum_{k=1}^{m} \eta_i \quad (2)$$

where η_i represents the S/N ratio at different levels.

$$\Delta F = F_{imax} - F_{imin} \quad (3)$$

3.3. Analysis of variance (ANOVA)

Analysis of Variance (ANOVA), also known as variance analysis, is a statistical method used to evaluate the significance of each control factor's contribution to improving experimental quality. Unlike Main Effects Analysis, which focuses on the impact of factors at different levels, ANOVA emphasizes quantifying the contribution of each factor to the overall system performance. This analysis is based on the F-distribution and begins with the calculation of the sum of squares (SS) and degrees of freedom (DF) for each control factor. These values are subsequently used to compute the variance ((Var)and the F-value, which determines the statistical significance of the factors.

(1) Total um of squares (SST):

$$SS_T = \sum (X_i - \overline{X})^2 = \sum_{n=1}^{n} {X_i}^2 - \frac{\left(\sum X_i\right)^2}{n} \quad (4)$$

where X_i is the value of each data point, $\overline{X}$ is the mean of all data points, and n is the number of experiments. This is used to measure the deviation of the data from the overall mean.

(2) Sum of squares for control factors (SSC)

$$SS_C = \sum_{j=1}^{k} n_j \left(\overline{X}_j - \overline{X}\right)^2 \quad (5)$$

where n_j is the sample size at the j level of the control factor, $\overline{X}_j$ overlin is the mean value at the j level, overline $\overline{X}$ is the overall mean of all data points, and k is the number of levels for the control factor. The main purpose is to distinguish the variation caused by the control factor settings from the variation caused by other random or non-controlled factors.

(3) Sum of squares for error (SSE)

$$SS_E = SS_T - SS_C \quad (6)$$

(4) Degree of freedom (DOF)

$$DOF_T = n \times r - 1 = \text{Total number} - 1 \quad (7)$$

$$DOF_F = Lv - 1 = \text{Level} - 1 \quad (8)$$

$$DOF_E = DOF_T - DOF_F \quad (9)$$

(5) Mean square, also referred to as variance, represents the ratio of the sum of squares (SS) to the degrees of freedom (DF). Since the calculation of the sum of squares is directly influenced by the experimental data, dividing it by its corresponding degrees of freedom helps mitigate this influence, yielding the mean square or variance. This approach ensures a normalized measure of variability, making the results more interpretable and robust.

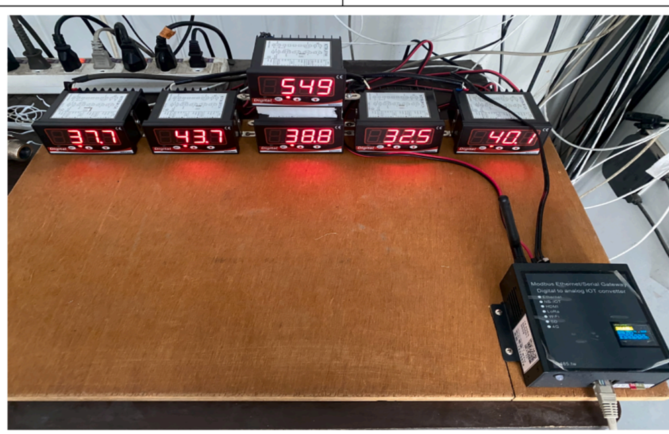
The formula for calculating the mean square is as follows:

E. Pump

F. Temperature sensor

G. Flow meter

H. Data logger



I. Sunlight, ambient temperature, and water temperature display

Fig. 1. (*continued*).

$$MS_f = \frac{SS_C}{DOF_F} \tag{10}$$

$$MS_e = \frac{SS_E}{DOF_E} \tag{11}$$

(6) F-value: The F-value is used to assess the relationship between the effect of a factor and the variation caused by random errors. The magnitude of the F-value directly reflects the degree of influence that the factor has on the system. Therefore, the importance of the factor can be determined based on the F-value. When the F-value is less than 1, it indicates that the effect of the factor is relatively small compared to the error; if the F-value is greater than 2, it signifies a noticeable effect of the factor; and if the F-value exceeds 4, it demonstrates a significant impact of the factor on the system. The F-value is calculated by dividing the mean square value of the factor by the mean square value of the error.

$$F = \frac{MS_f}{MS_e} \tag{12}$$

(7) Percent contribution (P): Percent contribution represents the proportion of each factor's sum of squares to the total sum of squares, indicating the relative effectiveness of the factor in reducing overall

Table 1
Detailed specifications of system modules.

Name	Parameter	Value
Flat plate collector (FPC)	Collector length (mm)	505
	Collector width (mm)	320
	Collector tube outer (mm)	25
	Collector tube inner (mm)	24
	Pipe number	9
	Collector plate material	Al
Heat storage tank	Tank volume (m^3)	0.04
	Tank height(mm)	500
	PCM pipe height(mm)	450
	PCM pipe number	12
	PCM pipe volume (m^3)	0.0037
	Heat exchanger outer (mm)	22
	Heat exchanger inner (mm)	21
	Heat exchanger outer (mm)	9
	Heat exchanger total volume (m^3)	0.003
Pyranometer	PYR2-420 – CLASS C/ 300 ~ 2900 nm/ 4 ~ 20 mA/ RS48510/Sensitivity:10 μV(W/m^2)	
Ambient temperature	**Galltec-mela** PC-ME series/relative humidity accuracy: (5…95 %rh at 10…40 °C) ± 2 % rh temperature accuracy: 0…1 V (−27…70 °C) ± 0.2 K、 0…10 V (−29…70 °C) ± 0.2、 K 4…20 mA (RC) ± 0.3 K	
Pump	**FORMOSA** RS-15/6GWS	
Temperature sensor	**PT100** – Platinum Resistance Temperature Sensor accuracy: **30 K to 305 K**: ±0.25 K **305 K to 400 K**: ±0.25 K **400 K to 475 K**: ±0.25 K **475 K to 500 K**: ±1.4 K **500 K to 670 K**: ±2.3 K	
Flow meter	**RP Series – Variable Area Flow Meter** 0.01~0.1 L/min up to 30~300 L/min/ Accuracy ± 2.5 % FSD	
Data logger	**BCT** TF 200S	

variation. This metric, referred to as the contribution, provides insight into the significance of each factor within the system. The primary purpose of this analysis is to ensure that all selected factors are meaningful contributors to system performance while confirming that no critical factors have been omitted. By quantifying the contribution of each factor, this analysis helps validate the experimental design and ensures the reliability of the results.

$$P = \frac{SS_C}{SS_T} \tag{13}$$

(8) P-value and confidence Level (CL): The P-value is a statistical measure used to express probability, widely applied in the analysis of experimental data. A smaller P-value indicates that the experimental results are statistically significant, highlighting the greater importance of the associated factor in the experiment. In contrast, a larger P-value suggests that the observed effects are more likely attributable to random errors rather than the direct influence of the control factor. This implies that the factor may have a smaller impact on the experimental outcomes. The confidence level (CL), typically expressed as a percentage, complements the P-value by indicating the probability that the results are not due to random chance, reinforcing the reliability of the analysis.

$$CL = (1 - P) \times 100\% \tag{14}$$

3.4. Grey relational analysis

Grey relational analysis is a quantitative method used to assess the similarities and differences between control factors by applying the Grey Relational Grade (GRG), which quantifies the degree of correlation between factors. In system analysis, particularly when quality characteristic data involve varying units of measurement or substantial differences in scale, it becomes essential to standardize the data to enable meaningful comparisons. This standardization process, integral to grey relational analysis, is referred to as grey relational generation.

(1) Grey Relational Generation: Grey relational generation involves standardizing data through an appropriate transformation method selected based on the target value of each data set. Typically, all data sequences are normalized to a range between 0 and 1, depending on the optimization objective: "larger-the-better," "smaller-the-better," or "nominal-the-best." As this study focuses on maximizing thermal storage efficiency and heat retention time, the "larger-the-better" calculation method was employed for normalization, ensuring consistency in data comparison and analysis.

$$x_i(k) = \frac{x_i(k) - \min, x_i(k)}{\max, x_i(k) - \min, x_i(k)} \tag{15}$$

where $x_i(k)$ represents the value after grey relational generation processing, $\max, x_i(k)$ is the maximum value in the original data, and $\max, x_i(k)$ is the minimum value in the original data.

(2) Grey Relational Coefficient: The grey relational coefficient is

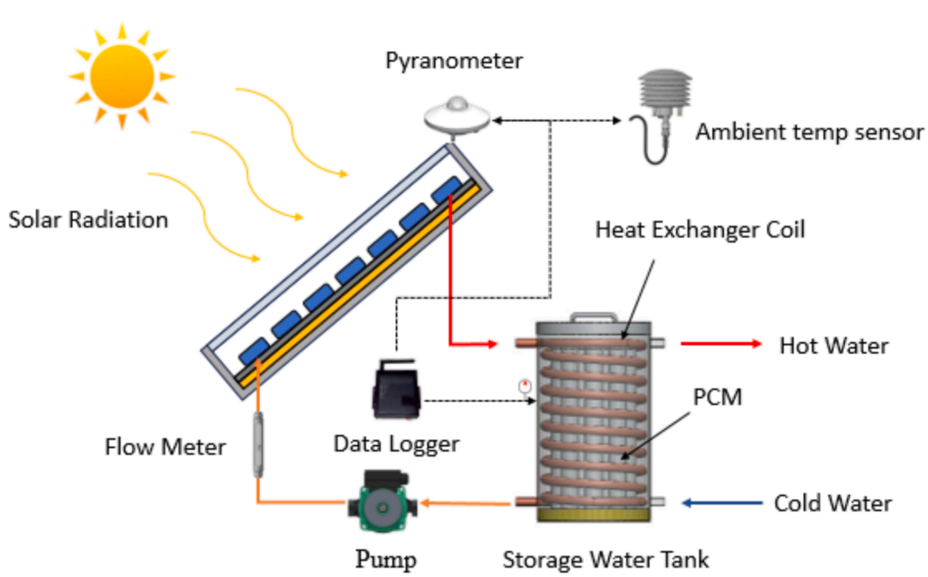


Fig. 2. Flow diagram of the integrated solar water heating system.

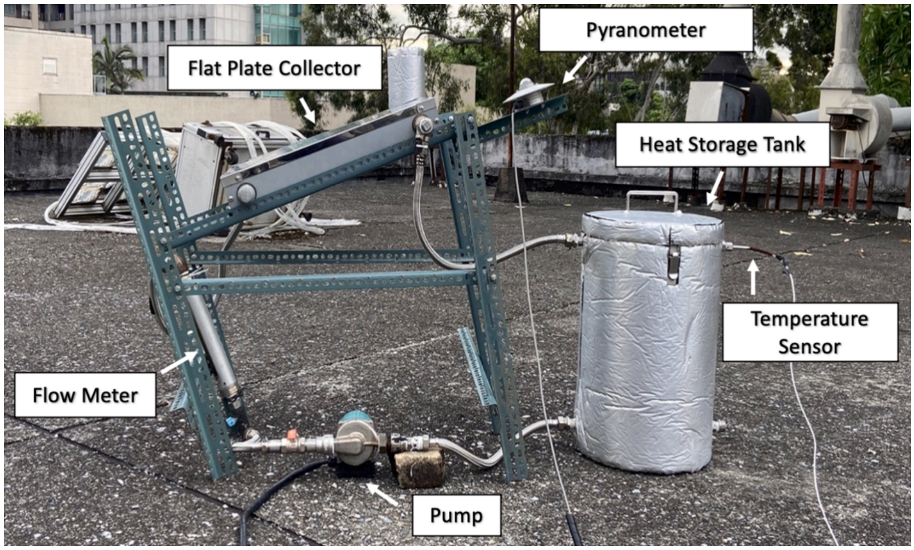


Fig. 3. Physical diagram of the integrated solar water heating system.

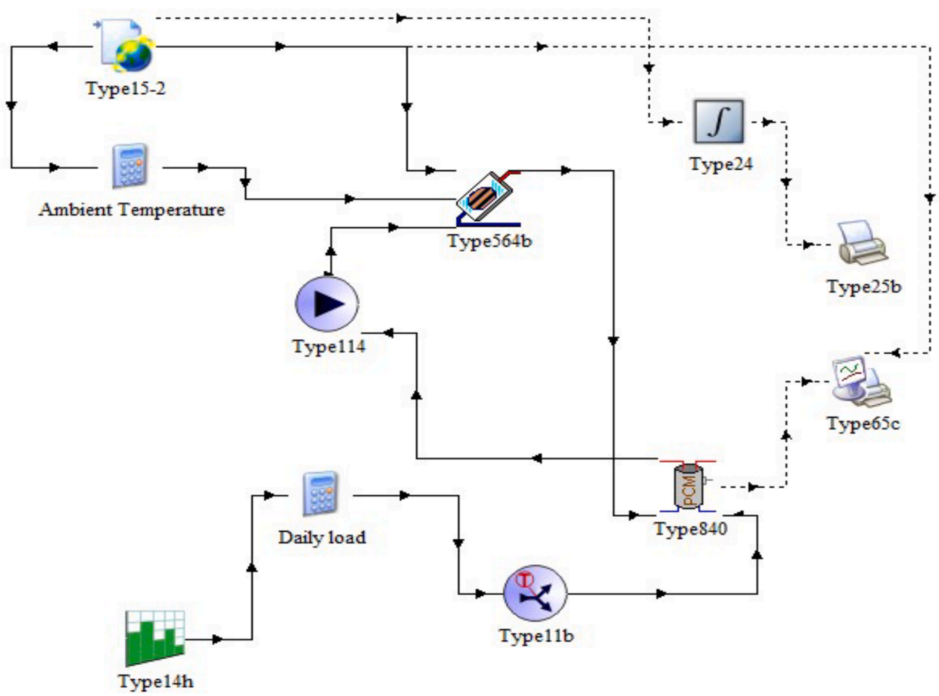


Fig. 4. System simulation architecture diagram.

calculated by selecting one sequence as the reference and treating the others as comparison sequences, a process referred to as local grey relational analysis. This coefficient quantifies the degree of correlation between the reference and comparison sequences. The calculation method for the grey relational coefficient is expressed as follows:

$$r_{0i}(k) = \frac{\min_i \min_k |x_0(k) - x_i(k)| + \zeta \max_i \max_k |x_0(k) - x_i(k)|}{|x_0(k) - x_i(k)| + \zeta \max_i \max_k |x_0(k) - x_i(k)|} \tag{16}$$

where $\min_i \min_k |x_0(k) - x_i(k)|$ refers to the minimum absolute difference between the reference sequence $x_0(k)$ and all comparison sequences $x_i(k)$ and $\max_i \max_k |x_0(k) - x_i(k)|$ is the maximum absolute difference among these differences. ζ is the distinguishing coefficient, typically set to 0.5.

(3) Grey relational grade calculation: The grey relational grade is determined by averaging the grey relational coefficients calculated for different sequences. This grade serves as an overall measure of the correlation between the reference sequence and the comparison sequences. The calculation method for the grey relational grade is expressed as follows:

$$r_i = \frac{1}{n} \sum_{k=1}^{n} w(k) \times \zeta_i(k) \tag{17}$$

The grey relational grade is used to evaluate the correlation between each set of experimental parameters and their ideal quality characteristic values. When the grey relational grade is high, it indicates that the set of parameters is closer to its ideal quality characteristic.

3.5. Heat transfer across the layers of a flat-plate solar collector

Heat transfer in flat-plate solar collectors occurs through conduction, convection, and radiation due to temperature differences, as governed by the second law of thermodynamics. Externally, heat is transferred via convection and radiation, influenced by ambient temperature, wind speed, and solar radiation. Internally, solar radiation penetrates the glass cover, heating the absorber plate and riser tubes through radiation, which in turn transfers heat to the working fluid by conduction. Convection within the air gap between the glass cover and absorber plate further distributes thermal energy. However, when the internal temperature exceeds the ambient temperature, heat loss to the environment occurs, reducing efficiency. To mitigate this, thermal insulation materials are employed to minimize heat dissipation, ensuring enhanced thermal performance and system efficiency. Fig. 5 illustrates these heat transfer processes within the collector's structural layers.

3.5.1. Assumptions for the model

To simplify the analysis of the flat-plate solar collector and accurately predict the dynamic temperature across its layers, a one-dimensional model with several assumptions was established to reduce unnecessary complexity and redundant calculations. The key assumptions are as follows:

(1) Uniform temperature distribution: The temperature within each layer of the collector is uniformly distributed, neglecting internal temperature gradients or non-uniformity.
(2) Perfect insulation at edges: The edges and sides of the flat-plate solar collector are assumed to have perfect thermal insulation, ignoring heat transfer mechanisms such as convection and radiation at the sides.
(3) Neglecting aluminum frame effects: The aluminum frame, primarily used for structural support, is excluded from the analysis due to its minimal area contribution and the increased complexity of including its thermal properties.
(4) Ignoring dust and dirt: The effects of dust and dirt accumulation on the collector surface are disregarded, assuming regular maintenance and cleaning to ensure optimal performance.
(5) Equal front and back temperatures: The ambient temperatures at the front and back surfaces of the collector are assumed to be identical, providing consistent environmental conditions for the analysis.

These assumptions simplify the heat transfer analysis while maintaining the focus on the key parameters influencing the system's performance. The corresponding heat transfer analysis based on these assumptions is illustrated in Fig. 6.

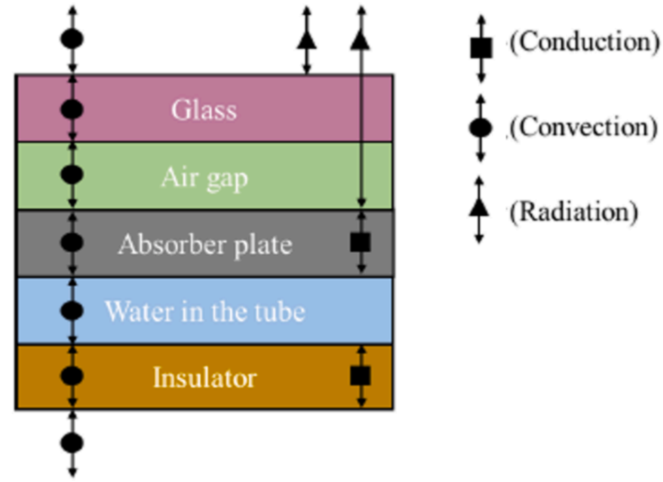


Fig. 6. Heat transfer mechanism analysis of the flat-plate solar collector.

3.5.2. Thermal model and equivalent circuit transformation

To simplify the complexity of thermal model analysis, the lumped capacitance method is employed, converting the thermal model into an equivalent electrical circuit. Each structural layer of the flat-plate solar collector is decomposed and represented as circuit components, facilitating the calculation and analysis of the system.

The performance of a flat-plate solar collector varies with the temperature of each component. Considering the effects of temperature and time changes in its equilibrium equations, a mathematical model is established. This model focuses on the primary components of the collector: (1) glass layer, (2) air gap, (3) absorber plate, (4) working fluid, and (5) insulation layer.

(1) Glass layer

The glass cover's temperature is influenced by ambient temperature, solar radiation, and wind speed on its upper surface, leading to convection and radiation. On the lower surface, heat transfer occurs through convection and radiation due to temperature differences with the air gap and absorber plate. Figs. 7 and 8 depict the thermal model and its equivalent circuit, respectively. The parameters of the heat transfer model for glass materials are shown in the Table 2.

After the thermal model was converted into an equivalent circuit, the heat balance equation for the glass layer was derived using circuit analysis (Kirchhoff's Current Law), as shown in Equation (18).

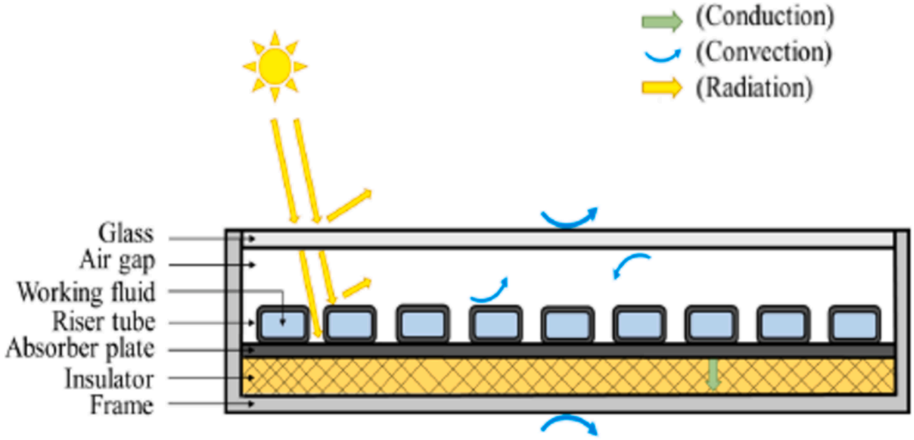


Fig. 5. Heat transfer analysis of flat-plate solar collectors.

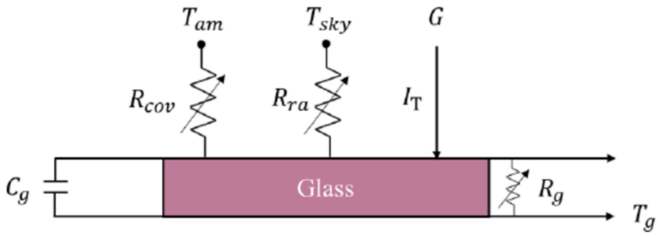


Fig. 7. Thermal model of the glass layer.

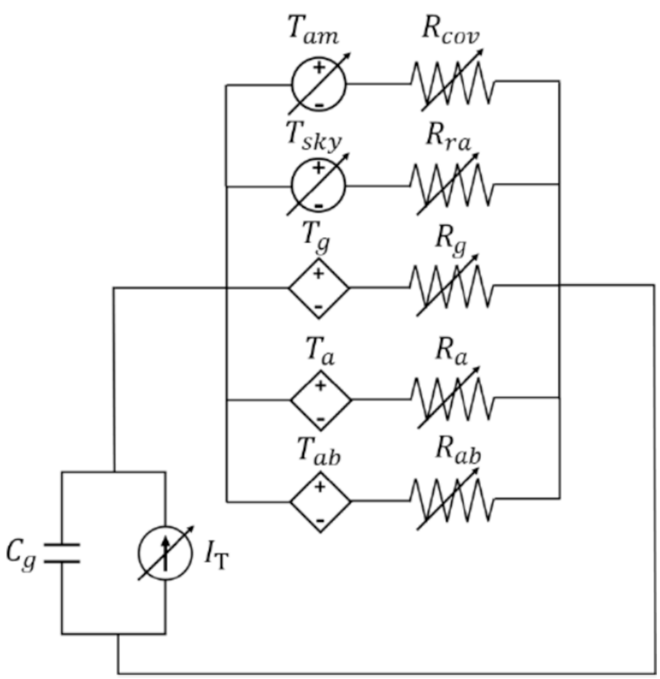


Fig. 8. Equivalent circuit of the glass layer.

Table 2
Parameters for the heat transfer model of the Glass Layer.

Parameter	Symbol	Unit	Value
Density of glass	ρ_g	kg/m^3	3000
Area of glass	A_g	m^2	0.1616
Thickness of glass	δ_g	m	0.003
Specific heat capacity of glass	c_g	J/kg K	500
Heat capacity of glass	C_g	J/K	7338.15
Absorptivity of glass	α_g		0.05
Emissivity of glass	ϵ_g		0.85
Emissivity of absorber plate	ϵ_{ab}		0.17
Stefan-Boltzmann constant	σ	W/m^2 K^4	5.6697×10^{-8}
Tilt angle	β	degree	23

$$Cg\frac{dTg}{dt} = IT + \frac{1}{Rra}(Tsky - Tg) + \frac{1}{Rcov}(Tam - Tg) + \frac{1}{Rra}(Tab - Tg) + \frac{1}{Rcov}(Ta - Tg) \tag{18}$$

Cg represents the thermal capacity of the glass layer, as defined in Equation (19).

$$Cg = \rho_g A_g \delta_g c_g \tag{19}$$

I_T represents the amount of solar radiation received by the glass layer, as defined in Equation (20).

$$I_T = \alpha_g G A_g \tag{20}$$

where α_g represents the absorptivity of the glass layer.

R_{ra} represents the radiative thermal resistance, which includes the radiation from the upper and lower surfaces of the glass layer. It is expressed as Equations (21) and (22).

$$R_{ra} = \sigma\varepsilon g(T_{sky} + T_g)(T_{sky}{}^2 + T_g{}^2) \tag{21}$$

$$R_{ra} = \frac{\sigma(T_{ab} + T_g)(T_{ab}{}^2 + T_g{}^2)}{\frac{1}{\varepsilon ab} + \frac{1}{\varepsilon g} - 1} \tag{22}$$

where σ is the Stefan-Boltzmann constant, εg represents the emissivity of the glass layer, and εab denotes the emissivity of the absorber plate.

T_{sky} represents the atmospheric temperature, as expressed in Equation (23).

$$T_{sky} = 0.0522T_{am}^{1.5} \tag{23}$$

represents the convective thermal resistance, which accounts for convection above and below the glass layer, as expressed in Equations (24) and (25).

$$R_{cov} = \frac{1}{h_{cov_{g-am}}A} \tag{24}$$

$$R_{cov} = \frac{1}{h_{cov_{g-a}}A} \tag{25}$$

where $h_{cov_{g-am}}$ represents the convective heat transfer coefficient between the glass and the ambient air, as expressed in Equation (26).

$h_{cov_{g-a}}$ represents the convective heat transfer coefficient between the glass and the air layer, as expressed in Equation (27).

$$h_{cov_{g-am}} = 5.7 + 3.8V_w \tag{26}$$

$$h_{cov_{g-a}} = N_{u_a}\frac{k_a}{\delta_a} \tag{27}$$

where N_{u_a} represents the Nusselt number for convection within the air layer between two structures, which is related to the Reynolds number as expressed in Equation (28).

$$N_{u_a} = 1 + 1.44\left[1 - \frac{1708}{R_a cos\beta}\right]^+ + \left[1 - \frac{1708sin1.8\beta^{1.6}}{R_a cos\beta}\right] + \left[\left(\frac{R_a cos\beta}{5830}\right)^{1/3} \Big/ -1\right]^+ \tag{28}$$

where the inclination angle (β) varies only between 0° and 75°. The Reynolds number (RaRaRa) in Equation (28) is calculated using Equation (29), while the volumetric expansion coefficient (β') is determined using Equation (30).

$$R_a = \frac{g\beta' \Delta T L^3 P_r}{v^2} \tag{29}$$

$$\frac{1}{\beta'} = \frac{(Tp + Tg)}{2} \tag{30}$$

where R_a is the Rayleigh number, g represents gravitational acceleration (m/s^2), L is the characteristic length, P_r is the Prandtl number, and v denotes the kinematic viscosity of the fluid.

(2) Air layer

The air layer's temperature is determined by the energy balance

Fig. 9. Thermal model of the air Layer.

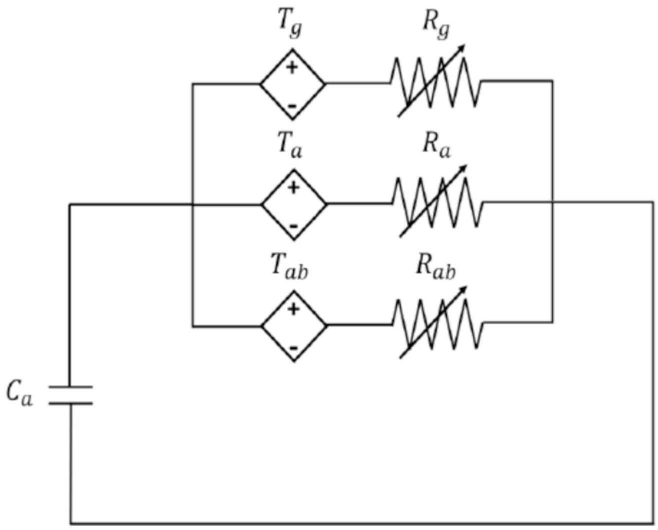


Fig. 10. Equivalent circuit of the air layer.

Table 3
Parameters of the air layer heat model.

Parameter	Symbol	Unit	Value
Density	ρ_a	kg/m^3	1.275
Area	$\boldsymbol{A_a}$	m^2	0.1616
Thickness	$\boldsymbol{\delta_a}$	m	0.0495
Specific Heat	$\boldsymbol{c_a}$	J/kg K	993
Heat Capacity	$\boldsymbol{C_a}$	J/K	10.13
Thermal Conductivity	k_a	W/m • K	0.026

between the glass and the absorber plate, with heat transfer primarily via convection. Fig. 9 shows the thermal model, and its equivalent circuit is presented in Fig. 10. All the parameters of air layer heat model are shown in the Table 3.

The heat balance equation is given in Equation (31).

$$C_a\frac{dT_a}{dt} = \frac{1}{Rcov}(Tg - T_a) + \frac{1}{Rcov}(Tab - T_a) \tag{31}$$

C_a represents the thermal capacity of the air layer, as shown in Equation (32).

$$Ca = \rho_a A_a \delta_a c_a \tag{32}$$

Rcov represents the convective thermal resistance, accounting for convection between the air layer and the upper glass as well as the lower absorber plate, as shown in Equation (33).

$$R_{cov} = \frac{1}{h_{cov_{g-a}}A} = \frac{1}{h_{cov_{ab-a}}A} \tag{33}$$

where $h_{cov_{g-a}}$ is the convective heat transfer coefficient between the glass and the air layer, and $h_{cov_{ab-a}}$ is the convective heat transfer coefficient between the absorber plate and the air layer. Both are expressed in Equation (34).

$$h_{cov_{g-a}} = h_{cov_{ab-a}} = N_{u_a}\frac{k_a}{\delta_a} \tag{34}$$

(3) Absorber plate

The absorber plate, the solar collector's key component, has the

Fig. 11. Thermal model of the absorber plate.

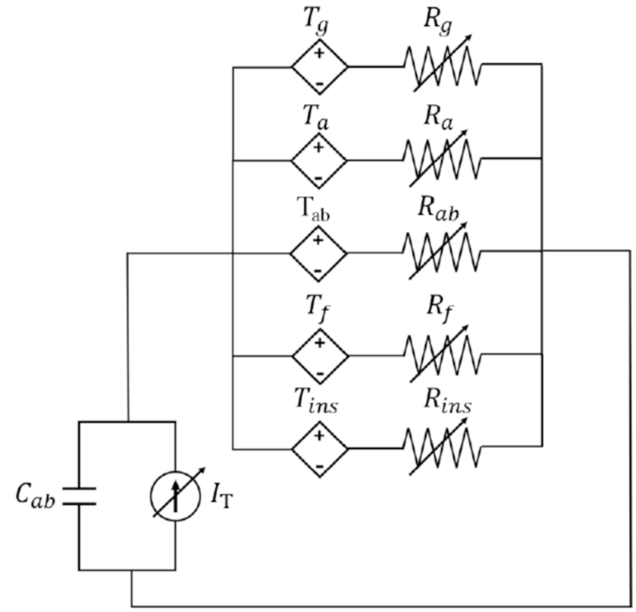


Fig. 12. Equivalent circuit of the absorber Plate.

Table 4
Thermal model parameters for the absorber plate.

Parameter	Symbol	Unit	Value
Density	ρ_{ab}	kg/m^3	2700
Area	A_{ab}	m^2	0.1616
Thickness	δ_{ab}	m	0.032
Specific heat	c_{ab}	J/kg K	896
Heat capacity	C_{ab}	J/K	12510.17
Absorptivity	α_{ab}		0.9
Height of cross-section	H	m	0.013
Width of cross-section	W	m	0.026

highest temperature, driven by energy balance between the glass and insulation layers. It undergoes solar radiation absorption, convective and radiative heat transfer on the upper side, and conductive heat transfer with the insulation below. Figs. 11 and 12 show its thermal model and equivalent circuit. The model's parameters are shown in the Table 4.

After converting the thermal model into an equivalent circuit, the heat balance equation for the absorber plate is derived using circuit analysis (Kirchhoff's Current Law) and is expressed in Equation (35).

$$C_{ab}\frac{dT_{ab}}{dt} = IT + \frac{1}{Rra}(Tg - T_{ab}) + \frac{1}{Rcov}(Ta - T_{ab}) + \frac{1}{R_{con}}(T_{ins} - T_{ab}) + \frac{1}{Rcov}(T_f - T_{ab}) \tag{35}$$

Cab represents the heat capacity of the absorber plate, as shown in Equation (36).

$$Cab = \rho_{ab} A_{ab} \delta_{ab} c_{ab} \tag{36}$$

I_T represents the thermal radiation received by the absorber plate, as expressed in Equation (37).

$$I_T = \alpha_{ab}\tau_g G A_{ab} \tag{37}$$

Where α_{ab} is the absorptivity of the absorber plate, and τ_g is the transmissivity of the glass layer.

R_{con} represents the conductive thermal resistance, specifically the heat conduction between the absorber plate and the insulation layer, as expressed in Equation (38).

$$R_{con} = \frac{1}{h_{con_{ins-ab}} A} \tag{38}$$

$h_{con_{ins-ab}}$ represents the convective heat transfer coefficient between the insulation layer and the absorber plate, as expressed in Equation (39).

$$h_{con_{ins-ab}} = \frac{k_{ins}}{\delta_{ins}} \tag{39}$$

represents the convective thermal resistance for both the upper and lower surfaces of the glass, as given by Equation (40).

$$R_{cov} = \frac{1}{h_{cov_{f-ab}} A} \tag{40}$$

The convective heat transfer coefficient between the glass and the ambient air, $h_{cov_{f-ab}}$, is given by Equation (41).

$$h_{cov_{f-ab}} = N_{u_f} \frac{k_f}{D_h} \tag{41}$$

The hydraulic diameter DhD_hDh, considering a rectangular cross-section, is expressed by Equation (42).

$$D_h = \frac{4HW}{2(H+W)} = \frac{2HW}{H+W} \tag{42}$$

where H represents the height of the cross-section, and W represents the width of the cross-section.

(4) Working fluid

The temperature of the working fluid is determined by the energy balance within the absorber plate region. After absorbing heat, the absorber plate transfers thermal energy to the cooler working fluid through convective heat transfer. Fig. 13 illustrates the thermal model of the working fluid, and its equivalent circuit representation is shown in Fig. 14 and paremeters of the thermal model for working fluid are described in the Table 5.

Through the thermal model and its equivalent circuit transformation, the heat balance equation for the working fluid is derived using circuit analysis (Kirchhoff's Current Law), as shown in Equation (43).

$$C_f \frac{dT_f}{dt} = \frac{1}{Rcov}(T_{ab} - T_f) + \dot{m}_f C_{pf}(T_{fo} - T_{fi}) \tag{43}$$

The heat capacity of the working fluid, C_f, is expressed as shown in Equation (44).

$$C_f = \rho_f A_f \delta_f c_f \tag{44}$$

The convective thermal resistance, R_{cov}, between the working fluid and the absorber plate is given by Equation (45).

$$R_{cov} = \frac{1}{h_{cov_{f-ab}} A} \tag{45}$$

The convective heat transfer coefficient, $h_{cov_{f-ab}}$, between the working fluid and the absorber plate is expressed in Equation (46).

$$h_{cov_{f-ab}} = N_{u_f} \frac{k_f}{D_h} \tag{46}$$

(5) Insulation layer

The insulation layer's temperature results from the absorber plate's energy balance, where heat is conducted from the plate to the cooler insulation layer below. Fig. 15 shows the thermal model, Table 6 shows the parameters of thermal model, and its equivalent circuit is illustrated in Fig. 16.

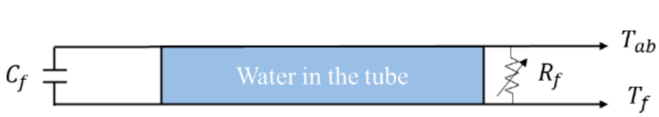


Fig. 13. Thermal model of the working fluid.

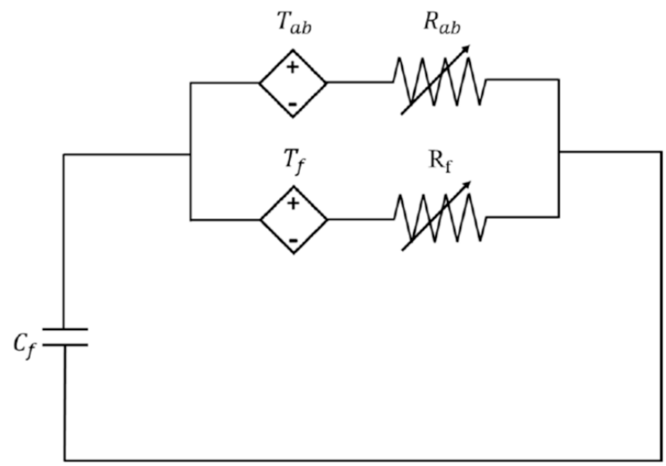


Fig. 14. Equivalent circuit of the working fluid.

Table 5
Parameters for the thermal model of the working fluid.

Parameter	Symbol	Unit	Value
Density	ρ_f	kg/m^3	1000
Cross-sectional area	A_f	m^2	0.006812
Thickness	δ_f	m	0.013
Specific heat	c_f	J/kg K	42,000
Heat capacity	C_f	J/K	371.94
Thermal conductivity	k_f	W/m • K	0.6

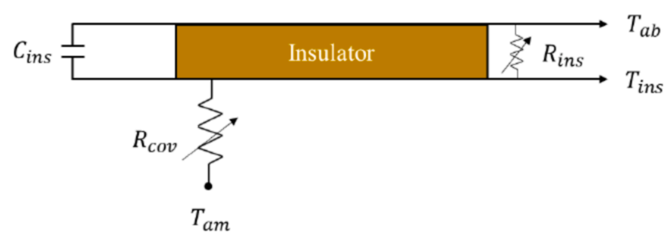


Fig. 15. Thermal model of the insulation layer.

Table 6
Insulation layer thermal model parameters.

Parameter	Symbol	Unit	Value
Density	ρ_{ins}	kg/m^3	20
Area	A_{ins}	m^2	0.1616
Thickness	δ_{ins}	m	0.025
Specific Heat	c_{ins}	J/kg K	1300
Heat Capacity	C_{ins}	J/K	105.04
Thermal Conductivity	k_{ins}	W/m • K	0.036

By converting the thermal model into its electrical circuit equivalent and applying circuit analysis (Kirchhoff's Current Law), the heat balance equation for the insulation layer is derived, as shown in Equation (47).

$$C_{ins} \frac{dT_{ins}}{dt} = \frac{1}{Rcon}(T_{ab} - T_{ins}) + \frac{1}{Rcov}(Tam - T_{ins}) \tag{47}$$

The thermal capacity of the insulation layer, C_{ins}, is expressed as shown in Equation (48).

$$C_{ins} = \rho_{ins} A_{ins} \delta_{ins} c_{ins} \tag{48}$$

The conductive thermal resistance, R_{con}, representing the heat conduction between the absorber plate and the insulation layer, is given by

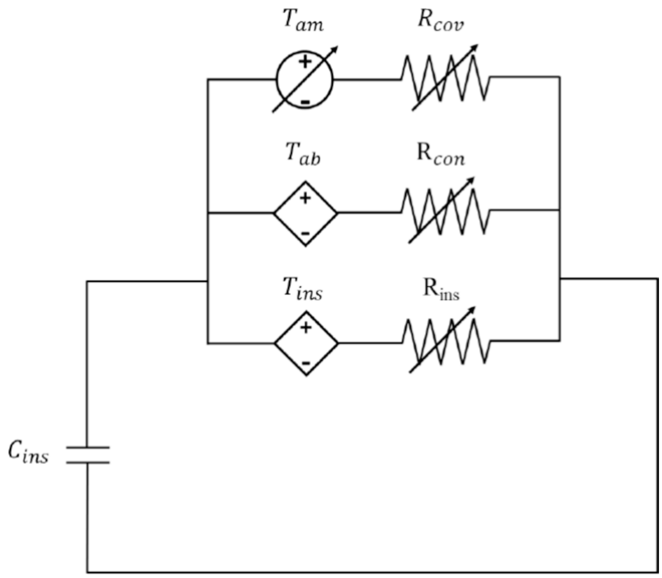


Fig. 16. Equivalent circuit of the insulation Layer.

Equation (49).

$$R_{con} = \frac{1}{h_{con_{ab-ins}} A} \tag{49}$$

The convective heat transfer coefficient, $h_{con_{ab-ins}}$, representing the convection between the insulation layer and the absorber plate, is defined by Equation (50).

$$h_{con_{ab-ins}} = \frac{k_{ins}}{\delta_{ins}} \tag{50}$$

The convective thermal resistance, R_{cov}, representing convection above and below the glass layer, is defined by Equation (51).

$$R_{cov} = \frac{1}{h_{cov_{am-ins}} A} \tag{51}$$

The convective heat transfer coefficient, $h_{con_{ab-ins}}$, representing convection between the insulation layer and ambient air, is defined by Equation (52)

$$h_{con_{ab-ins}} = 5.7 + 3.8 V_w \tag{52}$$

Through the integration of thermal models and equivalent circuit transformations, the heat transfer processes in flat-plate solar collectors were systematically analyzed. By decomposing the system into distinct layers—glass, air, absorber plate, working fluid, and insulation—this study captured the dynamics of conduction, convection, and radiation within each layer. Key findings highlight the significant influence of environmental factors such as wind speed and solar radiation, as well as the critical role of insulation in minimizing heat loss and enhancing efficiency. These insights provide a robust framework for optimizing the design and performance of flat-plate solar collectors, bridging theoretical understanding and practical applications.

4. Results and discussions

Improving the thermal performance of an integrated solar water heating system by addressing two critical parameters: thermal storage efficiency and heat retention time. The following section provides a detailed theoretical basis for these quality characteristics, including the calculation of thermal energy storage, latent heat contributions, and efficiency metrics. Improving the thermal performance of an integrated solar water heating system by addressing two critical parameters: thermal storage efficiency and heat retention time. The following section provides a detailed theoretical basis for these quality characteristics, Additionally, the analysis incorporates key experimental control factors and explains the rationale behind their parameter settings, such as phase change materials (PCMs), nanofluids, mass flow rate, and collector configurations. These factors were systematically adjusted and analyzed to achieve optimal system performance.

The aim of this study is to improve the thermal storage efficiency and heat retention time of the integrated solar water heating system. To achieve this, the following section outlines the theoretical basis of these two quality characteristics.

Thermal Storage Efficiency:

The thermal storage efficiency is determined by measuring the temperature increase of the system over a specified period. This typically involves calculating the change in the liquid temperature within the tank from its initial temperature to the final temperature. The stored thermal energy (Q_{store}) is calculated using the following formula:

$$Q_{store} = V\, C_p\, \Delta T = V\, C_p (T_f - T_i) \tag{53}$$

where,

V = Volume of the storage tank.
C_p = Specific heat capacity of the fluid (kJ/kg•°C).
ΔT = Temperature change in the storage tank.
T_i= Initial temperature of the storage tank (°C).
T_f = Final temperature of the storage tank (°C).

If phase change materials (PCMs) are added to the system, it is also necessary to calculate the latent heat (Q_{latent}) released or absorbed during the melting or solidification process of these materials, as calculated by the following formula:

$$Q_{latent} = m \bullet C_p dT(s) + mL + m \bullet C_p dT \tag{54}$$

where,

Q = Total latent heat (J).
m = Mass of the phase change material (kg).
Cp = Specific heat capacity of the phase change material (J/kg•°C).
L = Latent heat of fusion
ΔT = Difference between the final temperature and the initial temperature.

Calculate the sum of the thermal storage energy and latent heat to determine the total thermal storage energy (Q_{total}), as calculated by the following formula:

$$Q_{total} = Q_{store} + Q_{latent} \tag{55}$$

Finally, the total thermal storage energy calculated is compared with the total incident solar energy received by the solar collector. This comparison is performed by relating the total thermal storage energy to the product of the collector area and the solar radiation. The thermal storage efficiency is calculated using the following formula:

$$\eta = \left(\frac{Q_{total}}{A \bullet E}\right) \bullet 100\% \tag{56}$$

where,

A = Collector area (m^2).
E = Solar radiation (MJ/m^2).

Heat retention time is defined as the duration during which the storage tank maintains the water temperature above a specified minimum target temperature after sunset, without the aid of solar radiation. The calculation of heat retention time involves the following steps:

(1) Determine the Target Temperature: Set a minimum target temperature based on research objectives. In this experiment, the target temperature is set at 30 °C.
(2) Record Water Temperature at Sunset: Measure and record the water temperature in the storage tank at sunset, which serves as the starting point for tracking heat retention.

(3) Monitor Temperature Changes: Periodically measure the water temperature in the storage tank until it drops below the target temperature.
(4) Calculate Retention Time: Calculate the time elapsed from sunset until the water temperature falls below the target temperature.

For this study, based on literature and empirical data, the selected control factors include PCM material, PCM volume, number of PCM tubes, working fluid, mass flow rate, number of collector tubes, collector plate material, collector tilt angle, and collector azimuth angle. These factors are detailed in Table 7.

The following is an explanation of the selected levels for each control factor:

(1) PCM Material.

This study focuses on organic paraffin, the most commonly used phase change material (PCM) in solar water heating systems, to evaluate its effectiveness in enhancing thermal storage efficiency and stability. The control factor for PCM material has two levels: one includes the use of PCM material, while the other excludes it.

The paraffin selected for this study is RT35HC, with its heating and cooling curves analyzed using data provided by Rubitherm. As shown in Fig. 17, the minimal difference in enthalpy values between the heating and cooling processes results in nearly overlapping curves across the entire temperature range, demonstrating consistent thermal behavior.

(2) PCM volume.

Building on the research conducted by Cabeza et al. [37], which demonstrated that varying PCM volumes can significantly enhance energy density and prolong water temperature retention, three levels of PCM volume were selected for this study. The estimated daily water demand was set at a minimum of 30 L, with the initial PCM volume comprising 10 % of the total volume. Based on this baseline, the PCM volume was incrementally increased by 5 % and 10 %, resulting in three levels for this control factor: 10 %, 15 %, and 20 % of the total volume.

(3) Number of PCM tubes.

Research by Park et al. [50] highlighted that increasing the number of PCM tubes enhances the initial heat transfer rate and melting rate due to the expanded heat transfer area. This finding is particularly relevant when installation space is limited, as increasing the number of PCM tubes can effectively improve system efficiency. Based on this principle, this study added 2 and 4 additional PCM tubes to the baseline configuration of 12 tubes. Consequently, the levels selected for this control factor were 12, 14, and 16 PCM tubes.

(4) Working fluid type.

To investigate the influence of nanofluids on the thermal efficiency of the integrated solar water heating system, this study selected two nanofluids with superior thermal properties—aluminum oxide (Al2O3) and copper oxide (CuO)—in addition to the traditionally used water. These three working fluids form the levels for this control factor: water, (Al2O3), and CuO.

Metal oxide nanoparticles, particularly aluminum oxide and copper oxide, are commonly employed in solar water heating systems due to their excellent thermal properties, high heat transfer efficiency, and stability. The physical properties of the nanoparticles used in this study are provided in Table 8, offering further insight into their suitability for enhancing system performance.

In this experiment, copper oxide (CuO) and aluminum oxide (Al2O3) nanofluids were selected for their superior thermal properties. The density and specific heat capacity of these nanofluids, as utilized in this study, are detailed in Table 9.

(5) Mass flow rate.

Based on the findings of Moghadam et al. [51] and Prakasam et al. [28], the optimal mass flow rate range for solar water heating systems was identified as between 0.02 kg/s and 0.03 kg/s. Their research indicated that increasing the mass flow rate beyond 0.03 kg/s, particularly up to 0.05 kg/s, led to a decline in system performance. Accordingly, this study selected three levels of mass flow rate for investigation: 0.02 kg/s, 0.025 kg/s, and 0.03 kg/s.

(6) Number of collector tubes.

In this study, the flat-plate solar collector has an area of 0.15 m2 and a tube diameter of 25 mm. Due to the size constraints of the collector, the maximum allowable number of collector tubes is 11. The original configuration included 9 collector tubes. Accordingly, this study selected 9, 10, and 11 collector tubes as the levels for this control factor, ensuring compatibility with the collector's physical dimensions while enabling performance comparisons.

(7) Collector plate material.

Solar collector plates are typically constructed from aluminum, copper, or stainless steel. Among these, copper possesses the highest thermal conductivity, making it the most efficient in heat transfer, though it is also the most expensive. Aluminum, which has the second-highest thermal conductivity, is widely used due to its balance of performance and affordability. Stainless steel, while less thermally conductive than copper or aluminum, is valued for its durability and resistance to corrosion, making it ideal for long-term applications. These materials offer distinct trade-offs between thermal efficiency, cost, and longevity.

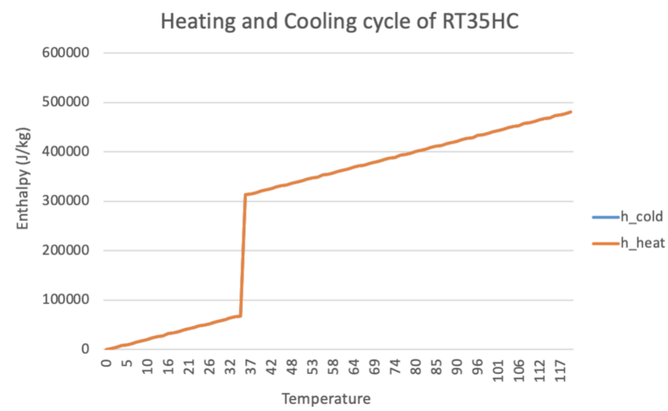


Fig. 17. Heating and cooling curves of paraffin RT35HC.

Table 7
Control Factors and their levels for the integrated solar water heating system.

Factor	Explanation	Level 1	Level 2	Level 3
A	PCM Material	No	Parffine	
B	PCM volume	10%	15%	20%
C	PCM tube	12	14	16
D	Work fluid	water	Al_2O_3	CuO
E	Mass flow rate (kg/s)	0.02	0.025	0.03
F	Collector tube number	9	10	11
G	Collector plate material	Cu	Al	M
H	Tilt angle (degrees)	20.4	22.4	24.4
I	Azimuth angle (degrees)	45	0	−45

Table 8
Specific physical properties of nanoparticles.

Name	Density	Specific heat capacity (kj/kg• ° C)	Particle size (mm)	Color
CuO	6310	0.531	35–40	Black
Al2O3	3970	0.765	30–55	White

Table 9
Thermal properties of nanofluids.

Nanofluid	Density(kg/m^3)	Capacity(kj/kg• ° C)
CuO	1210	3.41
Al2O3	1117	3.69

(8) Collector tilt angle.

Liu et al. [52] conducted an analysis to determine the optimal tilt angle for solar collectors in various regions of Taiwan. Their study revealed that the optimal tilt angle range for solar collectors in Taiwan lies between 20.2° and 22.4°, with 22.4° identified as the best angle for the Taipei area. Based on this finding, this study adjusted the tilt angle by ± 2°, selecting 20.4°, 22.4°, and 24.4° as the levels for this control factor to evaluate system performance across a representative range of tilt angles.

(9) Collector azimuth angle:

Selecting the appropriate azimuth angle is crucial for optimizing overall performance and maximizing solar radiation capture. Based on data from the Central Weather Bureau of the Ministry of Transportation and Communications regarding seasonal elevation and azimuth angles in Taiwan, this study selected −45°, 0°, and 45° as the levels for the collector azimuth angle control factor.

The experimental design was conducted systematically using the Taguchi method to define the levels of each control factor. The experimental data were then analyzed in detail through main effects analysis (MEA) and analysis of variance (ANOVA) to identify the optimal parameter combination for single quality characteristics, achieving the best performance for each specific metric. Finally, grey relational analysis (GRA) was applied to integrate multiple quality characteristics, enabling the determination of an optimal parameter combination that accounts for various quality attributes. This comprehensive approach ensures the practicality and robustness of the experimental results.

4.1. Validation of experimental and simulation data accuracy

To ensure the precision of the simulation analysis, a comparative evaluation was conducted between the simulation results and the empirical data obtained from the physical system. When the discrepancy between these two datasets is within 5 %, the simulation results are deemed reliable, thereby providing a robust foundation for subsequent optimization analyses. The TRNSYS model was configured and tested using the parameters derived from the physical system to ensure alignment between the simulation conditions and the actual operational environment. Subsequently, the simulation data generated by the TRNSYS model were compared against the experimental data from the physical system. The comparative outcomes are presented in Table 10.

According to Table 10, in the absence of PCM, the simulated data indicate an average thermal storage efficiency of 66.16 %, while the experimental data show an average value of 63.41 %, resulting in a discrepancy of 4.337 %. For heat retention time, the simulated data yield an average of 17.13 h, compared to 16.59 h from the experimental results, with a discrepancy of 3.195 %.

In the case of PCM utilization, since the experimental setup cannot directly measure the latent heat change of PCM, the thermal storage efficiency was calculated based on the observed temperature rise of the storage tank (sensible heat only). The simulated data were derived using the same approach, excluding the PCM latent heat contribution. The results indicate an average thermal storage efficiency of 62.43 % for the simulated data and 60.15 % for the experimental data, with a discrepancy of 3.791 %. Regarding heat retention time, the simulated data show an average of 18.33 h, while the experimental results average 17.66 h, with a discrepancy of 3.454 %.

In summary, under both PCM and non-PCM conditions, the discrepancies between the simulated and experimental data remain within a reasonable range of 5 %. This demonstrates that the simulation results are in strong agreement with the experimental observations, confirming their accuracy and reliability.

4.2. Thermal storage efficiency: Single quality optimization analysis

The experimental results for the thermal storage efficiency of the integrated solar water heating system are presented in Table 11. The corresponding main effects response table and graph, which detail the analysis of thermal storage efficiency, are provided in Table 12 and Fig. 18, respectively. These results highlight the influence of various control factors on the system's thermal performance, offering valuable insights for optimization.

By calculating the average S/N ratio for each level of the control

Table 11
Analysis results of thermal storage efficiency.

Run Group	1	2	3	4	5	Average	S/N
1	64.68	64.72	64.69	64.7	64.71	64.7	36.218
2	57.22	57.18	57.21	57.25	57.15	57.2	35.148
3	47.28	47.32	47.31	47.29	47.32	47.3	33.497
4	52.65	52.72	52.68	52.73	52.72	52.7	34.436
5	45.08	45.12	45.11	45.09	45.13	45.1	33.084
6	73.42	73.45	73.38	73.42	73.35	73.4	37.314
7	58.15	58.12	58.08	58.12	58.05	58.1	35.284
8	54.85	54.92	54.88	54.95	54.9	54.9	34.791
9	58.91	58.95	58.88	58.85	58.92	58.9	35.402
10	55.08	55.12	55.09	55.13	55.08	55.1	34.823
11	60.81	60.8	60.79	60.82	60.78	60.8	35.678
12	54.19	54.22	54.2	54.18	54.21	54.2	34.680
13	55.81	55.79	55.81	55.78	55.82	55.8	34.933
14	50.52	50.51	50.49	50.48	50.52	50.5	34.066
15	62.68	62.72	62.71	62.69	62.71	62.7	35.945
16	69.38	69.44	69.41	69.43	69.44	69.4	36.827
17	51.68	51.74	51.71	51.76	51.76	51.7	34.270
18	47.68	47.74	47.71	47.76	47.77	47.7	33.570
19	49.82	49.81	49.79	49.82	49.78	49.8	33.945
20	78.39	78.4	78.41	78.42	78.38	78.4	37.886
21	73.53	73.48	73.52	73.49	73.51	73.5	37.326
22	71.18	71.25	71.22	71.27	71.18	71.2	37.050
23	74.08	74.12	74.11	74.14	74.06	74.1	37.396
24	55.48	55.52	55.53	55.54	55.56	55.5	34.886
25	59.07	59.15	59.11	59.12	59.08	59.1	35.432
26	74.59	74.62	74.59	74.65	74.56	74.6	37.455
27	65.44	65.42	65.4	65.36	65.38	65.4	36.312
28	73.47	73.52	73.49	73.54	73.46	73.5	37.326
29	65.42	65.4	65.44	65.36	65.38	65.4	36.312
30	59.78	59.82	59.82	59.84	59.76	59.8	35.534
31	59.68	59.62	59.61	59.64	59.62	59.6	35.505
32	61.96	62.11	62.01	62.07	62.02	62	35.848
33	76.25	76.18	76.33	76.28	76.12	76.2	37.639
34	53.21	53.23	53.17	53.2	53.17	53.2	34.518
35	61.11	61.08	61.13	61.13	61.1	61.1	35.721
36	83.31	83.28	83.32	83.27	83.33	83.3	38.413
Average							35.679

Table 10
Comparison of experimental and simulation data.

	Without PCM				With PCM			
	Thermal storage efficiency (%)		Heat retention Time (h)		Thermal storage efficiency (%)		Heat retention time (h)	
	Experimental	Simulation	Experimental	Simulation	Experimental	Simulation	Experimental	Simulation
1	63.47	65.82	16.74	17.13	60.32	62.39	17.73	18.33
2	63.45	66.91	16.54	17.13	60.11	62.48	17.63	18.33
3	63.33	65.77	16.51	17.13	60.03	62.44	17.62	18.33
Average	63.41	66.16	16.59	17.13	60.15	62.43	17.66	18.33
Error (%)	4.337 %		3.195 %		3.791 %		3.454 %	

Table 12
Main effects response table for thermal storage efficiency.

	A	B	C	D	E	F	G	H	I
Lv 1	34.998	35.525	35.652	35.093	35.761	36.252	36.780	35.765	35.666
Lv 2	36.361	35.638	35.779	35.730	35.708	35.763	35.688	35.744	35.826
Lv 3		35.877	35.608	36.216	35.569	35.024	34.571	35.531	35.547
Max	36.361	35.877	35.779	36.216	35.761	36.252	36.780	35.765	35.826
Min	34.998	35.525	35.608	35.093	35.569	35.024	34.571	35.531	35.547
Diff	1.363	0.352	0.127	0.486	0.192	1.227	2.209	0.234	0.279
Rank	2	5	9	4	8	3	1	7	6

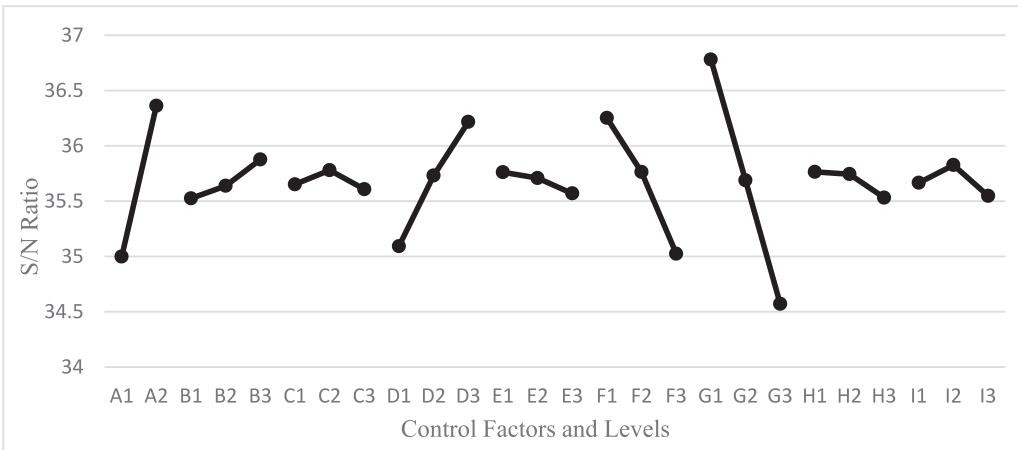


Fig. 18. Main effects response chart for thermal storage efficiency.

factors and compiling these averages into Table 12: Main Effects Response Table for Thermal Storage Efficiency, the impact of each control factor on thermal storage efficiency is clearly demonstrated. By comparing the average S/N ratios of different levels for each factor, the level with the highest average S/N ratio is selected as the optimal configuration, as it maximizes thermal storage efficiency while accounting for variability. This analysis not only highlights the importance of processing experimental data but also effectively quantifies the influence of each factor on system performance, ultimately identifying the optimal parameter configuration to enhance the performance of the integrated solar water heating system.

The main effects response table and chart reveal that the optimal factor levels for maximizing thermal storage efficiency are A2, B3, C2, D3, E1, F1, G1, H1, and I2. These correspond to the following configurations: the use of PCM material, 20 % PCM volume, 14 PCM tubes, CuO nanofluid as the working fluid, a mass flow rate of 0.02 kg/s•m^2, 9 collector tubes, copper as the collector plate material, a tilt angle of 20.4°, and an azimuth angle of 0° facing south. According to the main effects response table, the factor with the greatest influence on the thermal storage efficiency of the integrated solar water heating system is G (collector plate material), followed by A (PCM material), F (number of collector tubes), D (working fluid type), B (PCM volume), I (azimuth angle), H (tilt angle), E (mass flow rate), and C (number of PCM tubes).

As detailed in Table 13, each control factor significantly impacts thermal storage efficiency. Among these, PCM material demonstrates the most pronounced effect, followed by collector plate material, PCM volume, azimuth angle, tilt angle, and mass flow rate. These findings underscore the importance of optimizing material selection and geometric configuration to enhance system performance.

To ensure the accuracy and reliability of the single quality optimization results, a series of confirmation experiments is conducted to validate the effectiveness and reproducibility of the optimized parameter combination.

The validation process begins by calculating the overall mean of the Signal-to-Noise (S/N) ratios obtained from the 36 experiments. This is achieved by summing the S/N ratios of all experiments and dividing the total by 36. The calculation method is expressed as follows:

Table 13
Analysis of Variance (ANOVA) for Thermal Storage Efficiency of the Integrated Solar Water Heating System.

Factor	Degrees of Freedom (DOF)	Sum of Squares (SS)	Variance (Mean square, MS)	F-value	Contribution	P-value	Confidence Level
PCM material	1	16.72157	16.72157	440.520	25.51 %	0	100 %
PCM volume	2	0.77446	0.38723	10.201	1.18 %	0.001	99.9 %
PCM (No of tubes)	2	0.19070	0.09535	2.512	0.29 %	0.109	89.1 %
Working fluid	2	7.61675	3.80837	100.329	11.62 %	0	100 %
Mass flow rate	2	0.23649	0.11824	3.115	0.36 %	0.068	93.2 %
Number of collector tubes	2	9.16188	4.58094	120.682	13.98 %	0	100 %
Collector tube material	2	29.28081	14.64041	385.693	44.68 %	0	100 %
Inclination angle	2	0.40197	0.20098	5.295	0.61 %	0.015	98.5 %
Azimuth angle	2	0.47190	0.23595	6.216	0.72 %	0.008	99.2 %
error	18	0.68326	0.03796				
Combined error	18	0.68326	0.03796				
Total	35	65.53979					

$$\overline{\eta i} = \frac{1}{36}\sum_{i=1}^{36}\eta i = \frac{1}{36}(36.218 + 35.148 + 33.497 + \cdots + 36.218) = 35.679$$

Next, the overall mean and the optimal levels of the significant control factors are used to predict the optimized Signal-to-Noise (S/N) ratio. Based on Table 8, all control factors are identified as critical to thermal storage efficiency. The optimal level for each control factor is determined from Table 6, enabling the calculation of the predicted optimal S/N ratio. The calculation method is expressed as follows:

$$\begin{aligned}\widehat{\mathrm{SN}} =& \overline{\eta i} + (A_2 - \overline{\eta i}) + (B_3 - \overline{\eta i}) + (C_2 - \overline{\eta i}) + (D_3 - \overline{\eta i}) + (E_1 - \overline{\eta i}) \\ &+ (F_1 - \overline{\eta i}) + (G_1 - \overline{\eta i}) + (H_1 - \overline{\eta i}) + (I_2 - \overline{\eta i}) = 39.179\end{aligned}$$

Finally, the CI value for the 95 % confidence interval is calculated. The calculation method is as follows:

$$I = \sqrt{F_{a;1,V_2} \times MS_e \times \left[\frac{1}{n_{eff}} + \frac{1}{r}\right]} = \sqrt{4.41 \times 0.03796 \times 0.7} = 0.34231$$

The confidence interval for the optimized thermal storage efficiency is calculated to be between 38.8376 and 39.5222. If the Signal-to-Noise (S/N) ratio obtained from the confirmation experiments falls within this range, it confirms that the experiment is both highly accurate and reproducible.

As shown in Table 14, the S/N ratio calculated from five confirmation experiments is 39.294, which lies within the 95 % confidence interval. This result indicates that the experimental findings are reliable and demonstrate good reproducibility, validating the effectiveness of the optimized parameter combination.

4.3. *Heat retention time: Single quality optimization analysis*

The experimental results for the heat retention time of the integrated solar water heating system are presented in Table 15. The corresponding main effects response table and chart, which detail the analysis of heat retention time, are provided in Table 16 and Fig. 19, respectively. These results offer insights into the influence of various control factors on the system's heat retention performance and guide the identification of optimal parameter combinations.

The average S/N ratios for each level of the control factors were calculated and compiled into Table 16, providing a clear representation of the influence of each factor on thermal storage efficiency.

The main effects response table and chart identify the optimal factor levels for maximizing heat retention time as A2, B1, C2, D3, E1, F1, G1, H2, and I2. These levels correspond to the following configurations: the use of PCM material, 10 % PCM volume, 14 PCM tubes, CuO nanofluid as the working fluid, a mass flow rate of 0.02 kg/s, 9 collector tubes, copper as the collector plate material, a tilt angle of 22.4°, and an azimuth angle of 0° facing south.According to the main effects response table, the factor with the greatest influence on heat retention time is G (collector plate material), followed by D (working fluid type), F (number of collector tubes), A (PCM material), I (azimuth angle), H (tilt angle), C (number of PCM tubes), E (mass flow rate), and B (PCM volume).

As shown in Table 17, the collector plate material exhibits the most significant impact on heat retention time, followed by the working fluid, the number of collector tubes, and the PCM material. In contrast, PCM volume, the number of PCM tubes, mass flow rate, tilt angle, and azimuth angle show minimal influence on heat retention time. Consequently, these five control factors were grouped into the error term for further analysis.

Table 15
Analysis results of heat retention time.

Frequency Groups	1	2	3	4	5	Mean	S/N ratio
1	16.31	16.3	16.3	16.33	16.3	16.3	24.244
2	10.81	10.81	10.8	10.83	10.8	10.8	20.668
3	0	0	0	0	0	0	0.000
4	5.26	5.26	5.3	5.33	5.32	5.3	14.486
5	0	0	0	0	0	0	0.000
6	22.2	22.23	22.2	22.21	22.18	22.2	26.927
7	12.33	12.28	12.3	12.33	12.31	12.3	21.798
8	7.66	7.71	7.73	7.68	7.71	7.7	17.730
9	12.46	12.51	12.48	12.46	12.51	12.5	21.938
10	10.1	10.06	10.06	10.08	10.11	10.1	20.086
11	12.76	12.8	12.83	12.81	12.78	12.8	22.144
12	7.53	7.48	7.46	7.52	7.51	7.5	17.501
13	10.36	10.41	10.43	10.38	10.4	10.4	20.341
14	0	0	0	0	0	0	0.000
15	17.4	17.38	17.41	17.36	17.43	17.4	24.811
16	20.06	20.13	20.08	20.13	20.1	20.1	26.064
17	4.53	4.46	4.48	4.5	4.51	4.5	13.064
18	0	0	0	0	0	0	0.000
19	0	0	0	0	0	0	0.000
20	27.06	27.13	27.11	27.08	27.12	27.1	28.659
21	18.58	18.61	18.56	18.6	18.63	18.6	25.390
22	18.06	18.13	18.1	18.11	18.08	18.1	25.154
23	22.26	22.3	22.28	22.28	22.31	22.3	26.966
24	0	0	0	0	0	0	0.000
25	3.56	3.63	3.6	3.58	3.61	3.6	11.126
26	22.98	23.01	23	23.03	22.96	23	27.235
27	10.71	10.68	10.73	10.7	10.66	10.7	20.588
28	21.56	21.63	21.6	21.58	21.61	21.6	26.689
29	11.01	10.98	11	11.03	10.96	11	20.828
30	3.76	3.83	3.8	3.78	3.82	3.8	11.596
31	7.8	7.78	7.79	7.82	7.76	7.8	17.842
32	10.21	10.18	10.22	10.2	10.19	10.2	20.172
33	19.33	19.41	19.58	19.5	19.66	19.5	25.801
34	3.1	3.06	3.12	3.08	3.1	3.1	9.827
35	8.16	8.22	8.2	8.22	8.18	8.2	18.276
36	28.38	28.41	28.4	28.36	28.41	28.4	29.066
Mean value							17.695

To ensure the accuracy and reliability of the single quality optimization results, confirmation experiments are conducted to validate the effectiveness and reproducibility of the optimized parameter combination.

The validation process begins by calculating the overall mean of the Signal-to-Noise (S/N) ratios obtained from the 36 experiments. This is achieved by summing the S/N ratios of all experiments and dividing the total by 36. The calculation method is expressed as follows:

$$\overline{\eta i} = \frac{1}{36}\sum_{i=1}^{36}\eta i = \frac{1}{36}(24.244 + 20.668 + 0 + \cdots + 29.066) = 17.695$$

Next, the overall mean and the optimal levels of the significant control factors are used to predict the optimized Signal-to-Noise (S/N) ratio. Based on Table 11, four control factors are identified as critical to heat retention time. The optimal level for each of these factors is determined from Table 10, enabling the calculation of the predicted optimal S/N ratio. The calculation method is expressed as follows:

$$\widehat{\mathrm{SN}} = \overline{\eta i} + (A_2 - \overline{\eta i}) + (D_3 - \overline{\eta i}) + (F_1 - \overline{\eta i}) + (G_1 - \overline{\eta i}) = 33.809$$

Finally, the CI value for the 95 % confidence interval is calculated. The calculation method is as follows:

Table 14
Confirmation Experiment Results for Thermal Storage Efficiency of the Integrated Solar Water Heating System.

Primary Control Factors	1	2	3	4	5	Mean	S/N Ratio
A_2、B_3、C_2、D_3、E_1、F_1、G_1、H_1、I_2	92.17	92.23	92.21	92.22	92.19	92.2	39.294

Table 16
Main effects response table for heat retention time.

	A	B	C	D	E	F	G	H	I
Level 1	16.211	18.138	18.110	13.172	18.491	20.917	25.259	18.151	17.431
Level 2	19.179	17.979	18.263	18.374	16.975	18.585	19.772	18.276	18.949
Level 3		16.968	16.712	21.540	17.619	13.583	8.055	16.658	16.705
Max	19.179	18.138	18.263	21.540	18.491	20.917	25.259	18.276	18.949
Min	16.211	16.968	16.712	13.172	16.975	13.583	8.055	16.658	16.705
Diff	2.967	1.170	1.551	8.368	1.516	7.334	17.204	1.618	2.243
Rank	4	9	7	2	8	3	1	6	5

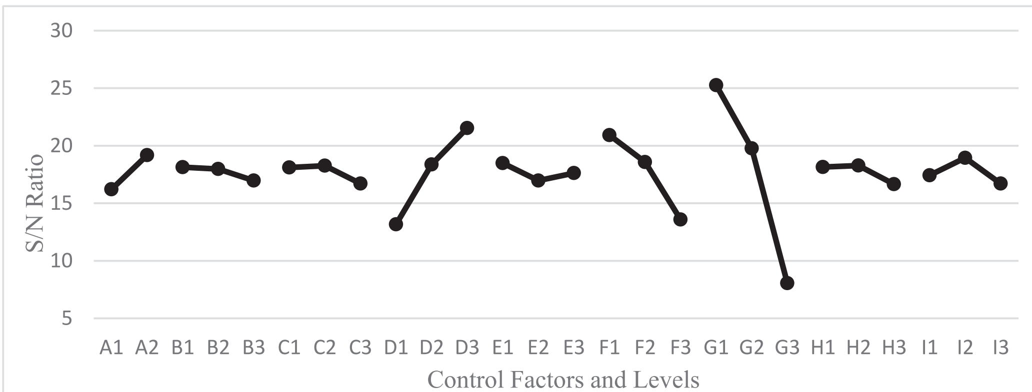


Fig. 19. Main effects response chart for heat retention time.

Table 17
Analysis of variance for heat retention time.

Factor	Degrees of freedom (DOF)	Sum of squares (SS)	Variance (mean square, MS)	F-value	Contribution	P-value	Confidence level
PCM material	1	79.24580	79.24580	4.752	2.56 %	0.042	95.8 %
PCM volume	2	9.65940	4.82970	0.290	0.31 %	0.751	24.9 %
PCM (No of tubes)	2	17.53974	8.76987	0.526	0.57 %	0.598	40.2 %
Working fluid	2	428.4361	214.2180	12.84	13.86 %	0	100.0 %
Mass flow rate	2	13.88591	6.94296	0.416	0.45 %	0.666	33.4 %
Number of collector tubes	2	336.9901	168.4950	10.10	10.9 %	0.001	99.9 %
Collector tube material	2	1853.466	926.7334	55.57	59.98 %	0	100.0 %
Inclination angle	2	19.45315	9.72657	0.583	0.63 %	0.568	43.2 %
Azimuth angle	2	31.44773	15.72386	0.943	1.02 %	0.409	59.1 %
error	18	300.1825	16.67681				
Combined error	28	392.1684	14.00602				
Total	35	3090.307					

$$CI = \sqrt{F_{\alpha;1,V_2} \times MS_e \times \left[\frac{1}{n_{eff}} + \frac{1}{r}\right]} = \sqrt{4.2 \times 14.00602 \times 0.422} = 4.98$$

The confidence interval for the optimized heat retention time is calculated to be between 28.8291 and 38.7891. If the Signal-to-Noise (S/N) ratio obtained from the confirmation experiments lies within this range, it indicates that the experiment is both highly accurate and reproducible.

As shown in Table 18, the S/N ratio obtained from five confirmation experiments is 29.425, which falls within the 95 % confidence interval. This result confirms that the experimental findings are reliable and demonstrate good reproducibility, validating the effectiveness of the optimized parameter combination.

4.4. Multi-quality optimization analysis for the integrated solar water heating system

Practical applications require optimizing multiple performance aspects, such as thermal storage efficiency and heat retention time. To address this, Grey Relational Analysis (GRA) was used to identify the best parameter configuration.

The process analyzed 36 experimental results, using Signal-to-Noise (S/N) ratios to compare thermal storage efficiency and heat retention time. Maximum and minimum values were identified and normalized, as shown in Table 19, forming the basis for determining the optimal configuration. This approach ensures the system achieves balanced and effective performance in real-world applications.

To maximize both thermal storage efficiency and heat retention time, the "larger-the-better" approach was used during the analysis. In this

Table 18
Confirmation experiment results for heat retention time.

Primary control factor	1	2	3	4	5	Mean value	S/N ratio (db)
A_2、D_3、F_1、G_1	29.62	29.57	29.59	29.61	29.58	29.6	29.425

Table 19
S/N ratios for thermal storage efficiency and heat retention time.

	S/N ratio for thermal storage efficiency (db)	S/N ratio for thermal storage time (db)
1	36.218	24.244
2	35.148	20.668
3	33.497	0.000
4	34.436	14.486
5	33.084	0.000
6	37.314	26.927
7	35.284	21.798
8	34.791	17.730
9	35.402	21.938
10	34.823	20.086
11	35.678	22.144
12	34.680	17.501
13	34.933	20.341
14	34.066	0.000
15	35.945	24.811
16	36.827	26.064
17	34.270	13.064
18	33.570	0.000
19	33.945	0.000
20	37.886	28.659
21	37.326	25.390
22	37.050	25.154
23	37.396	26.966
24	34.886	0.000
25	35.432	11.126
26	37.455	27.235
27	36.312	20.588
28	37.326	26.689
29	36.312	20.828
30	35.534	11.596
31	35.505	17.842
32	35.848	20.172
33	37.639	25.801
34	34.518	9.827
35	35.721	18.276
36	38.413	29.066
Max	38.413	29.066
Min	33.084	0
Diff	5.329	29.066

Table 20
Grey relational generation for thermal storage efficiency and heat retention time.

	Thermal storage efficiency	Thermal storage duration
X0	1	1
X1	0.58816619	0.83408264
X2	0.38736099	0.71107873
X3	0.07762494	0
X4	0.25381643	0.4983601
X5	0	0
X6	0.7937882	0.92639922
X7	0.41280529	0.74994245
X8	0.32047246	0.60997698
X9	0.43509372	0.75476238
X10	0.32639907	0.69105395
X11	0.48683825	0.76184958
X12	0.29955795	0.60211259
X13	0.34697411	0.6998008
X14	0.18431763	0
X15	0.53699038	0.85359774
X16	0.7024581	0.89670379
X17	0.22259295	0.44946279
X18	0.09134979	0
X19	0.16156809	0
X20	0.90119304	0.98599822
X21	0.79600714	0.87352709
X22	0.74419108	0.86538409
X23	0.80925775	0.92774227
X24	0.33818802	0
X25	0.44061852	0.3827809
X26	0.82021822	0.93697836
X27	0.60570473	0.7082989
X28	0.79600714	0.9182116
X29	0.60570473	0.71656199
X30	0.45980917	0.39893778
X31	0.45434915	0.61383289
X32	0.51869234	0.6939981
X33	0.85480447	0.88764765
X34	0.26920668	0.3380964
X35	0.49486032	0.62877749
X36	1	1
Max. Value	1	1
Min. Value	0	0

method, the highest values from the two sets of Signal-to-Noise (S/N) ratios were chosen as the reference points (labeled as X0), representing the ideal performance. The remaining data were compared against these reference points to assess how closely each setup approached the ideal. The results of these comparisons are shown in Table 20.

After completing the grey relational generation, the next step involves calculating the difference between each comparison sequence and the reference sequence X_0. The calculated differences quantify the deviation of each sequence from the reference, providing the basis for subsequent analysis. The results are presented in Table 21.

Finally, the difference sequences are substituted into the grey relational coefficient formula to calculate the grey relational grade values for thermal storage efficiency and heat retention time. The average of each set of grey relational grades is computed to obtain the final grey relational grade value. These results are summarized in Table 22.

To determine the best overall performance, an analysis was conducted on the grey relational grades, which represent the combined results for thermal storage efficiency and heat retention time. This analysis identifies how each control factor influences the system's overall performance and helps pinpoint the optimal settings for maximum efficiency. The results, including the recommended optimal levels, are clearly summarized in Table 23 and Fig. 20.

According to the main effects response table and chart, the optimal factor levels for multi-quality optimization are A2, B3, C2, D3, E1, F1, G1, H2, and I2. These levels correspond to the following configurations: the use of PCM material, 20 % PCM volume, 14 PCM tubes, CuO nanofluid as the working fluid, a mass flow rate of 0.02 kg/s, 9 collector tubes, copper as the collector plate material, a tilt angle of 22.4°, and an azimuth angle of 0° facing south.The main effects response table further reveals that the factor with the greatest influence on the thermal storage efficiency of the integrated solar water heating system is G (collector plate material), followed by D (working fluid type), F (number of collector tubes), A (PCM material), B (PCM volume), H (tilt angle), E (mass flow rate), C (number of PCM tubes), and I (azimuth angle).The results of the confirmation experiments for thermal storage efficiency and heat retention time are detailed in Table 24 and Table 25, respectively.

After five confirmation experiments, the thermal storage efficiency reached 94.2 %, and the heat retention time was 31.7 h. The Signal-to-Noise (S/N) ratio for thermal storage efficiency was 39.481, while the S/N ratio for heat retention time was 30.021. Both values fall within the 95 % confidence interval obtained from the single quality optimization process, providing strong evidence of the reliability and reproducibility of the optimized parameters derived through Grey Relational Analysis.

According to the main effects plot in Fig. 8, the configuration utilizing CuO nanofluid as the working fluid, 9 collector tubes, copper collector plates, a mass flow rate of 0.02 kg/s, a tilt angle of 22.4°, and a due south azimuth angle demonstrated significant improvements in system performance. The following sections provide a detailed discussion of each parameter:

(1) Working fluid: CuO nanofluid.

The experimental results demonstrated that CuO nanofluid exhibited superior thermal performance compared to Al_2O_3 nanofluid and water. This enhanced performance is primarily attributed to the higher thermal conductivity of CuO nanofluid, which exceeds that of Al_2O_3 nanofluid, with water showing the lowest thermal conductivity among the three.

Table 21
Difference values for thermal storage efficiency and heat retention time.

	Thermal storage efficiency	Thermal storage duration
X0	0	0
X1	0.41183381	0.16591736
X2	0.61263901	0.28892127
X3	0.92237506	1
X4	0.74618357	0.5016399
X5	1	1
X6	0.2062118	0.07360078
X7	0.58719471	0.25005755
X8	0.67952754	0.39002302
X9	0.56490628	0.24523762
X10	0.67360093	0.30894605
X11	0.51316175	0.23815042
X12	0.70044205	0.39788741
X13	0.65302589	0.3001992
X14	0.81568237	1
X15	0.46300962	0.14640226
X16	0.2975419	0.10329621
X17	0.77740705	0.55053721
X18	0.90865021	1
X19	0.83843191	1
X20	0.09880696	0.01400178
X21	0.20399286	0.12647291
X22	0.25580892	0.13461591
X23	0.19074225	0.07225773
X24	0.66181198	1
X25	0.55938148	0.6172191
X26	0.17978178	0.06302164
X27	0.39429527	0.2917011
X28	0.20399286	0.0817884
X29	0.39429527	0.28343801
X30	0.54019083	0.60106222
X31	0.54565085	0.38616711
X32	0.48130766	0.3060019
X33	0.14519553	0.11235235
X34	0.73079332	0.6619036
X35	0.50513968	0.37122251
X36	0	0

Table 22
Grey relational grade values for thermal storage efficiency and heat retention time.

	Thermal storage efficiency	Thermal storage duration	Mean value
X0	1	1	1
X1	0.54834554	0.75084392	0.64959473
X2	0.44938205	0.6337768	0.54157943
X3	0.35152472	0.33333333	0.34242903
X4	0.401225	0.49918139	0.45020319
X5	0.33333333	0.33333333	0.33333333
X6	0.70800289	0.8716864	0.78984465
X7	0.45989922	0.66661551	0.56325737
X8	0.42389854	0.56178322	0.49284088
X9	0.46952489	0.67092695	0.57022592
X10	0.4260392	0.61808819	0.52206369
X11	0.49350461	0.67736871	0.58543666
X12	0.41651323	0.55686269	0.48668796
X13	0.43364161	0.62484441	0.52924301
X14	0.38003094	0.33333333	0.35668213
X15	0.51920561	0.77351215	0.64635888
X16	0.62692631	0.82878027	0.72785329
X17	0.39141791	0.47594697	0.43368244
X18	0.35494972	0.33333333	0.34414153
X19	0.37357149	0.33333333	0.35345241
X20	0.83499363	0.97275927	0.90387645
X21	0.71023448	0.79811911	0.7541768
X22	0.66154287	0.78787814	0.72471051
X23	0.72385901	0.8737322	0.7987956
X24	0.43036224	0.33333333	0.38184778
X25	0.47197352	0.44753979	0.45975665
X26	0.73553016	0.88806533	0.81179775
X27	0.55909946	0.63155148	0.59532547
X28	0.71023448	0.85941898	0.78482673
X29	0.55909946	0.63821259	0.59865602
X30	0.48068103	0.45410695	0.46739399
X31	0.47817108	0.56422767	0.52119938
X32	0.5095242	0.62034593	0.56493507
X33	0.77495887	0.81652336	0.79574112
X34	0.40624205	0.4303283	0.41828517
X35	0.4974433	0.5739062	0.53567475
X36	1	1	1

(2) Number of Collector Tubes: 9 Tubes.

(3) The experimental results indicated that, for a fixed collector surface area, increasing the number of tubes reduces the spacing between them. This closer spacing leads to greater thermal interference, which in turn decreases the system's overall thermal efficiency. Therefore, maintaining 9 collector tubes provided an optimal balance between spacing and performance.

(4) Collector tube material: copper.

(5) Copper, known for its exceptionally high thermal conductivity, exhibited superior performance in the experiments. Compared to aluminum and stainless steel, copper transfers absorbed solar energy to the working fluid more efficiently, resulting in a significant enhancement of the system's overall thermal efficiency.

(6) Mass flow rate: 0.02 kg/s.

(7) The experimental results revealed that a lower mass flow rate of 0.02 kg/s allows the working fluid to fully absorb heat, preventing insufficient heat absorption caused by excessively rapid flow rates. Furthermore, a lower mass flow rate reduces the energy consumption of the pump, thereby decreasing operating costs and improving overall system efficiency.

(8) Tilt angle: 22.4 degrees.

(9) The experimental results showed that a tilt angle of 22.4 degrees maximized the amount of solar radiation received, outperforming the angles of 20.4 degrees and 24.4 degrees. While 20.4 degrees performed slightly below 22.4 degrees, 24.4 degrees received the least solar radiation, resulting in lower system efficiency.

(10) Azimuth angle: due south.

(11) The experimental results demonstrated that orienting the collector due south ensured maximum solar radiation exposure throughout the day. This configuration outperformed southeast orientation, which ranked second, and southwest orientation, which received the least solar radiation.

From the main effect plots in Figs. 18, 19, and 20, it is evident that both thermal storage efficiency and heat retention time improved significantly when PCM was utilized compared to scenarios without PCM. This demonstrates the substantial impact of PCM on enhancing the overall performance of the integrated solar water heating system.

Based on the multi-quality optimization experimental results, a PCM volume of 20 % provided an adequate amount of phase change material to effectively absorb and release heat, while 14 PCM tubes offered sufficient surface area to facilitate efficient heat transfer without significantly increasing the system's thermal resistance. This configuration achieved the optimal balance by enhancing thermal storage efficiency and extending heat retention time, all while minimizing excessive costs and system weight.

4.5. Comparative analysis of optimized results

Finally, the optimized research results were compared with findings from recent studies conducted over the past few years. The comparative analysis is presented in Table 26, highlighting the advancements achieved in this study relative to prior research.

According to the literature, Modi (2020), Gupta (2021) and García-Rincón(2024) reported efficiencies of 52.72 %, 50.27 % and 85 % respectively, in solar water heating systems using nanofluids. In contrast, studies by Mahfuz (2014), Feliński (2016), Itamar (2022), Bernal (2022) and El-Fakharany (2024) demonstrated significantly higher efficiencies for solar water heating systems employing phase change materials (PCMs), achieving 77.41 %, 79 %, 76.08 %, 77.85 %

Table 23
Main effects analysis of grey relational grade.

	A	B	C	D	E	F	G	H	I
Lv 1	0.5203	0.5587	0.5686	0.4856	0.5956	0.6422	0.7301	0.5871	0.5796
Lv 2	0.6372	0.5798	0.5956	0.5933	0.5832	0.5877	0.5805	0.5941	0.5808
Lv 3		0.5978	0.5722	0.6574	0.5575	0.5065	0.4259	0.5552	0.5759
Max	0.6372	0.5978	0.5956	0.6574	0.5956	0.6422	0.7301	0.5941	0.5808
Min	0.5203	0.5587	0.5686	0.4856	0.5575	0.5056	0.4259	0.5552	0.5759
Diff	0.1169	0.0391	0.0270	0.1718	0.0381	0.1357	0.3041	0.0389	0.0013
Rank	4	5	8	2	7	3	1	6	9

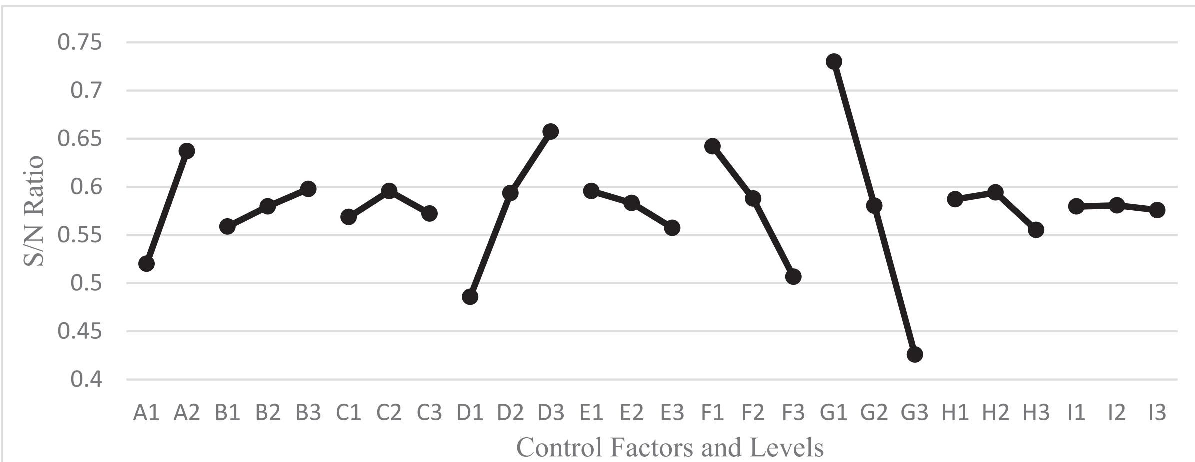


Fig. 20. Main effects plot of grey relational grade.

Table 24
Multi-quality confirmation results for thermal storage efficiency.

Primary control factor	1	2	3	4	5	Mean value	S/N ratio
A_2、B_3、C_2、D_3、E_1、F_1、G_1、H_2、I_2	94.14	94.16	94.2	94.22	94.18	94.2	39.481

Table 25
Multi-quality confirmation results for heat retention time).

Primary control factor	1	2	3	4	5	Mean value	S/N ratio
A_2、B_3、C_2、D_3、E_1、F_1、G_1、H_2、I_2	31.71	31.69	31.72	31.71	31.66	31.7	30.021

and 64.53 %, respectively. However, no prior research has combined nanofluids and PCMs in solar water heating systems.Further review of the literature reveals that the combination of nanofluids and PCMs is commonly applied in PV/T systems, significantly improving their performance. For instance, Liu (2023) demonstrated that this combination achieved a notable efficiency of 64.76 % in PV/T systems. To address this gap, To address this gap, we combined nanofluids and PCMs in solar water heating systems and optimized the parameter configurations to enhance system performance. The final experimental results indicate that the combined use of nanofluids and PCMs in solar water heating systems achieves an efficiency of 94.2 %, significantly surpassing the efficiencies reported for systems using nanofluids or PCMs individually. This highlights the potential of integrating these advanced materials to optimize solar water heating system performance.

5. Conclusions

The primary objective of this study was to optimize the parameters of a solar water heating system by integrating nanofluids, phase change materials (PCMs), and flat-plate collectors to achieve maximum thermal storage efficiency and extended heat retention time. This research not only provides novel theoretical insights but also demonstrates the practical potential of these innovations, offering a feasible and scalable solution to meet the growing global demand for energy-efficient technologies.

To achieve this, the Taguchi method was employed to systematically design experiments, focusing on nine critical control factors: the type of PCM material, PCM volume, number of PCM tubes, working fluid type (e.g., water or nanofluids), mass flow rate, number of collector tubes, collector plate material, tilt angle, and azimuth angle. A total of 36 experiments were conducted, systematically planned using the Taguchi orthogonal array to efficiently evaluate the impact of each parameter on system performance.

After completing the experiments, data analysis was conducted using Signal-to-Noise (S/N) ratios and main effects analysis to assess the influence of the control factors. Analysis of variance (ANOVA) was performed to identify statistically significant factors contributing to performance enhancement. To optimize both thermal storage efficiency and heat retention time simultaneously, Grey Relational Analysis (GRA) was applied, successfully identifying the optimal parameter configuration that balances these two quality characteristics.

This study demonstrates the effectiveness of a systematic

Table 26
Comparison of optimized results between this study and recent literature.

No.	Reference	Research Topic	Efficiency (%)
1	Mahfuz (2014) [53]	PCM-SHWS	77.41 %
2	Feliński (2016) [54]	PCM-SHWS (ETC)	79 %
3	Modi (2020) [55]	Nanofluid-SHWS (FPC)	52.72 %
4	Gupta (2021) [56]	Nanofluid-SHWS (FPC)	50.27 %
5	Bernal (2022) [57]	PCM-SHWS	76.08 %
6	Liu (2023) [44]	Nanofluid-PCM-PVT	64.76 %
7	García-Rincón and Flores-Prieto (2024) [24]	Nanofluid-SHWS (FPC)	85 %
8	Yousif (2024) [42]	Nanofluid-PCM-CPC-SHWS	77.1 %
9	El-Fakharany (2024) [22]	PCM-Integrated SAH	64.53 %
10	Optimization Results of the Present Study	Nanofluid-PCM-SHWS	94.2 %

optimization approach in significantly improving the performance of solar water heating systems. The results highlight that these optimized configurations not only improve energy utilization efficiency and extend heat retention time but are also highly suitable for practical applications in residential and commercial water heating systems. In the context of the global transition toward renewable energy and sustainable development, this research provides critical theoretical support and practical reference for advancing solar thermal technologies that are both efficient and reliable.

The key findings and practical implications are as follows:

(1) Single-quality optimization results:
- Thermal storage efficiency: The optimized thermal storage efficiency reached 92.2 %. The optimal parameter configuration included the use of PCM material, a PCM volume of 20 %, 14 PCM tubes, CuO nanofluid as the working fluid, a mass flow rate of 0.02 kg/s, 9 collector tubes, copper collector plates, a tilt angle of 20.4°, and an azimuth angle of 0° facing south. Compared to the non-optimized system, this represents an improvement of 26.04 %.
- Heat retention time: The optimized heat retention time was 29.6 h. The corresponding optimal parameters included PCM material, a PCM volume of 10 %, 14 PCM tubes, CuO nanofluid as the working fluid, a mass flow rate of 0.02 kg/s, 9 collector tubes, copper collector plates, a tilt angle of 22.4°, and an azimuth angle of 0° facing south. This configuration extended the heat retention time by 11.33 h compared to the non-optimized system.

(2) Multi-quality optimization results:

The optimized thermal storage efficiency was 94.2 %, and the heat retention time reached 31.7 h. The optimal parameter configuration included PCM material, a PCM volume of 20 %, 14 PCM tubes, CuO nanofluid as the working fluid, a mass flow rate of 0.02 kg/s, 9 collector tubes, copper collector plates, a tilt angle of 22.4°, and an azimuth angle of 0° facing south. Compared to the non-optimized system, these optimizations resulted in a 28 % improvement in thermal storage efficiency and an extension of heat retention time by 14.6 h.

These findings not only validate the effectiveness of systematic optimization techniques like the Taguchi method and Grey Relational Analysis in enhancing the performance of solar water heating systems but also underscore the potential of integrating advanced materials and multi-objective optimization methods to address the challenges of energy efficiency and sustainability. By achieving significant improvements in thermal storage efficiency and heat retention time, this research bridges the gap between theoretical advancements and practical applications, providing a scalable solution for residential and commercial energy systems. Furthermore, this study serves as a critical step toward advancing renewable energy technologies, offering valuable insights and actionable strategies to support global sustainability goals and the transition to a low-carbon future.

CRediT authorship contribution statement

Guan-Rong Chen: Visualization, Software, Methodology, Investigation. **Ting-Wei Liao:** Formal analysis, Data curation, Conceptualization. **Chien-Chun Hsieh:** Resources, Formal analysis. **Jagadish Barman:** Writing – review & editing, Validation. **Chao-Yang Huang:** Validation, Investigation, Data curation. **Chung-Feng Jeffrey Kuo:** Validation, Supervision, Funding acquisition, Conceptualization.

Funding

The research was supported by the National Science and Technology Council of the Republic of China under the grant No: NSTC 112-2622-E-011-028 and NSTC 113-2622-E-011-024.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The financial support provided by Energy Administration, Ministry of Economic Affairs, Taiwan, R.O.C. (Contract No: 114-S0102) is gratefully acknowledged.

Data availability

Data will be made available on request.

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